The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

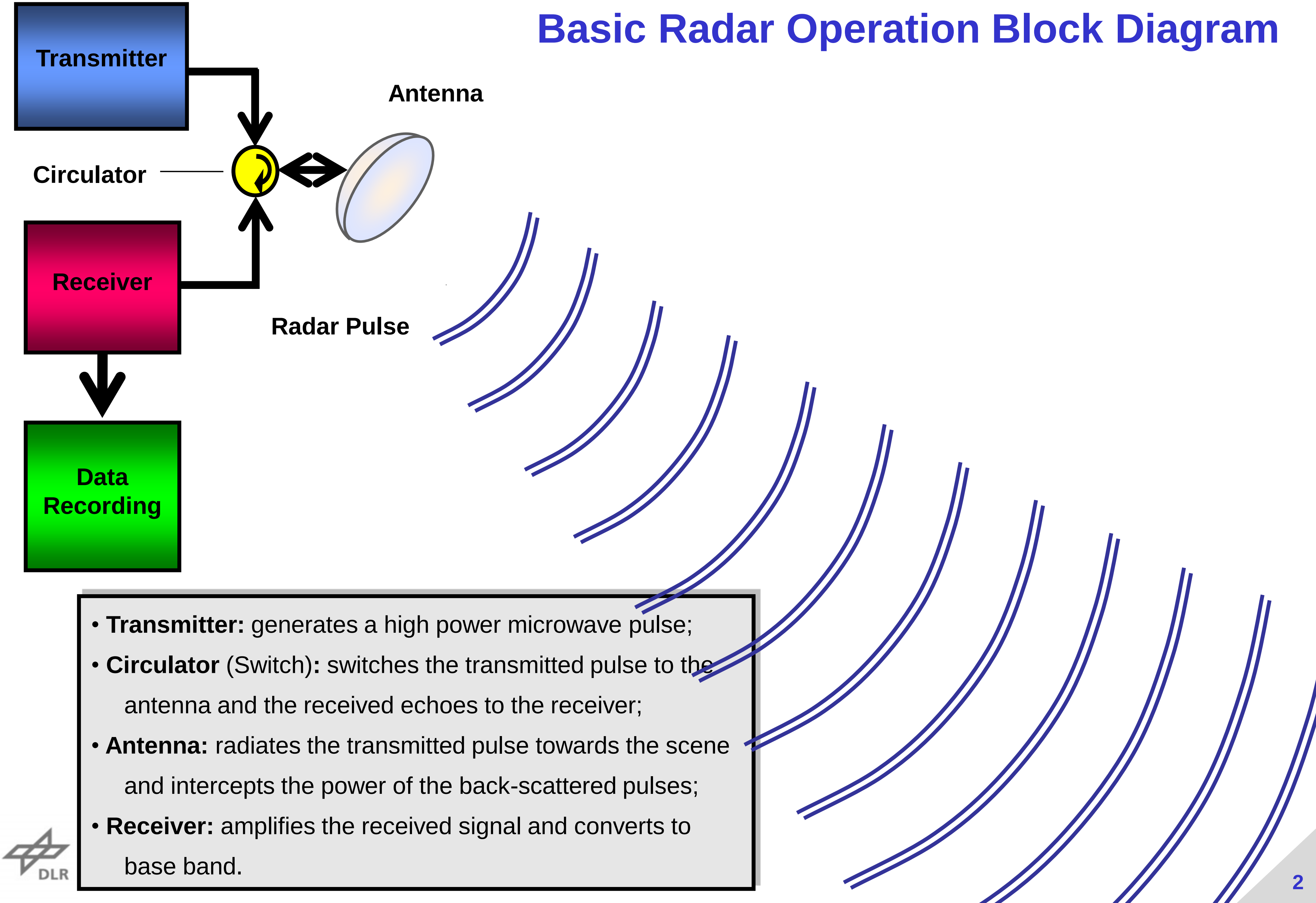
Part 1: Range Resolution

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

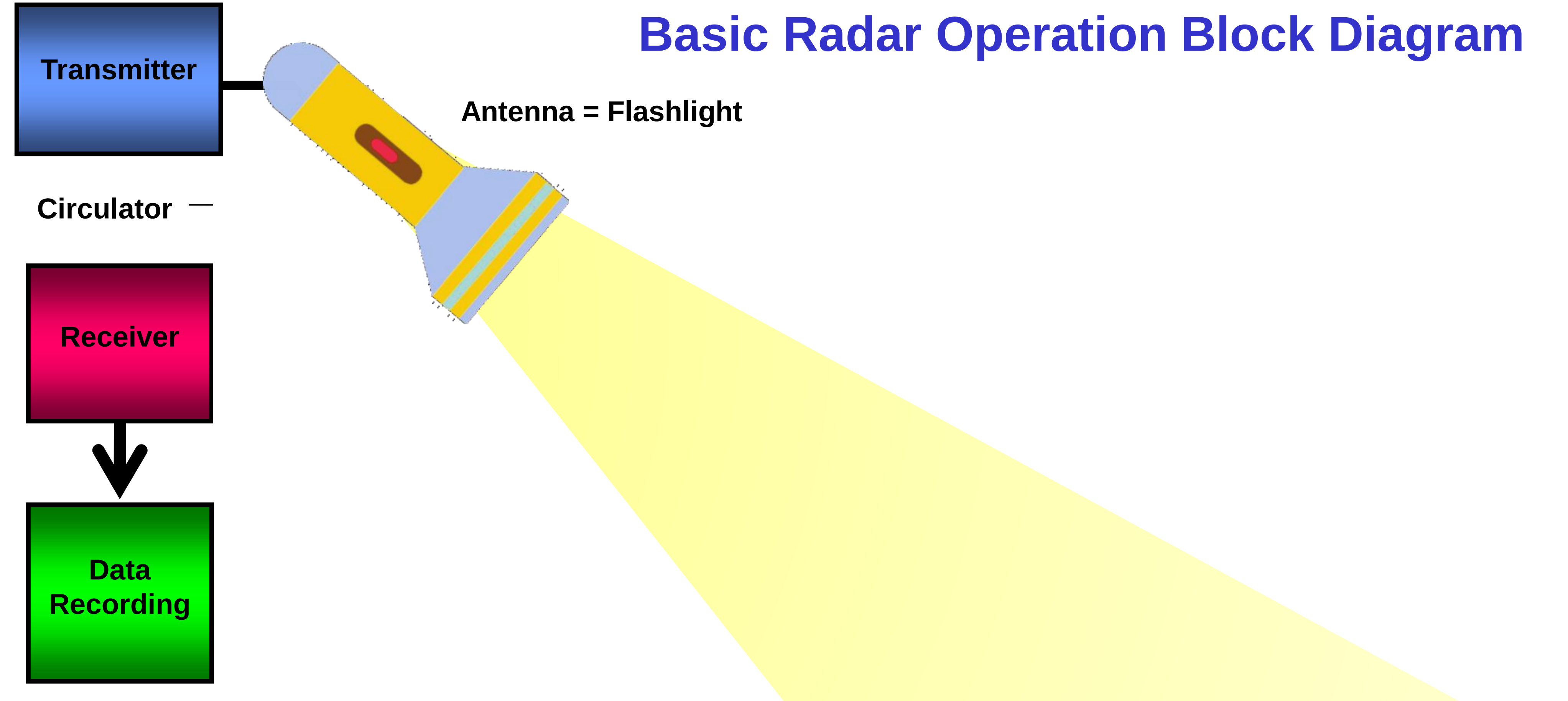
Email: kostas.papathanassiou@dlr.de, matteo.pardini@dlr.de, islam.mansour@dlr.de

Basic Radar Operation Block Diagram



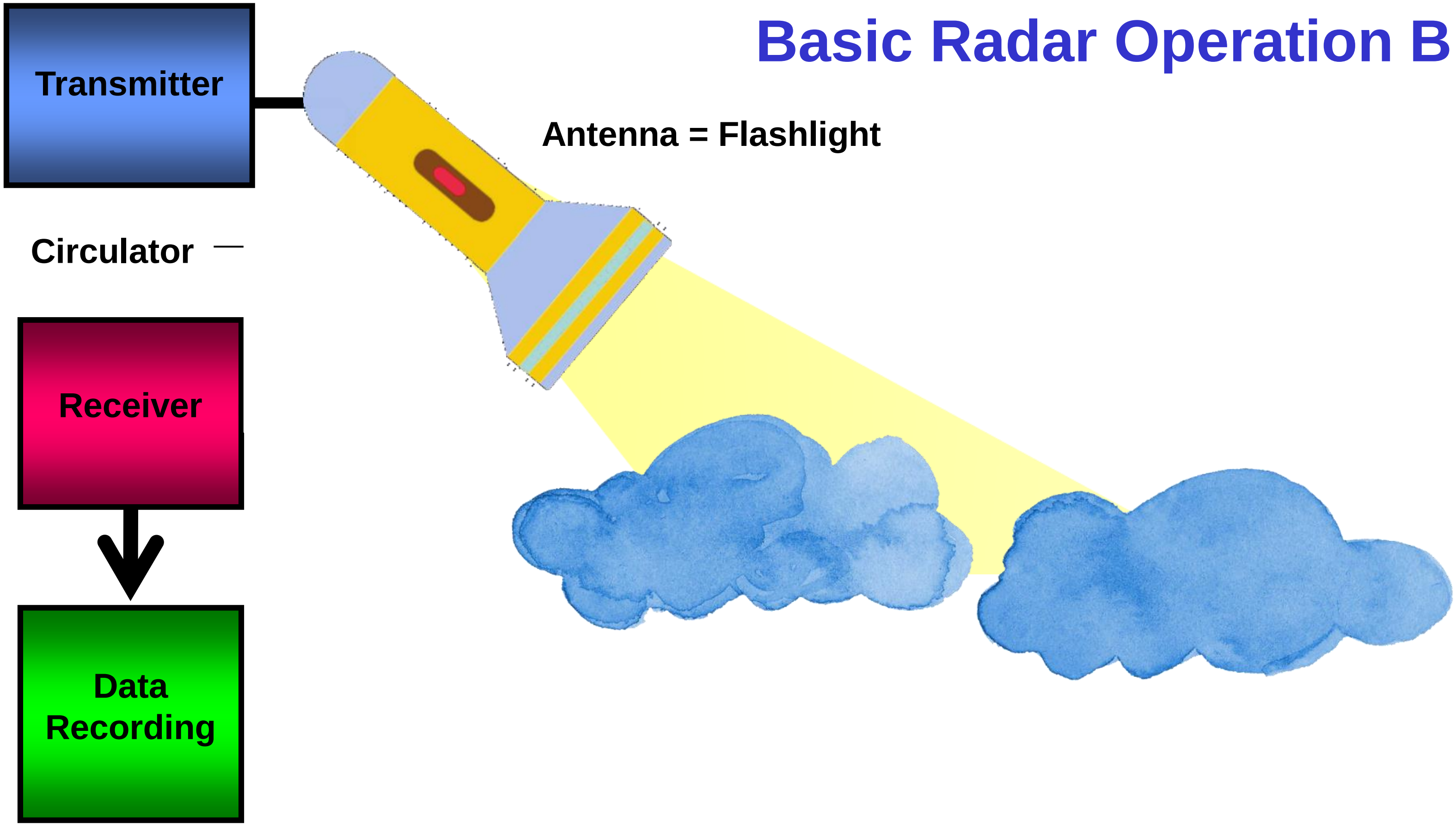
- **Transmitter:** generates a high power microwave pulse;
- **Circulator** (Switch)**:** switches the transmitted pulse to the antenna and the received echoes to the receiver;
- **Antenna:** radiates the transmitted pulse towards the scene and intercepts the power of the back-scattered pulses;
- **Receiver:** amplifies the received signal and converts to base band.

Basic Radar Operation Block Diagram



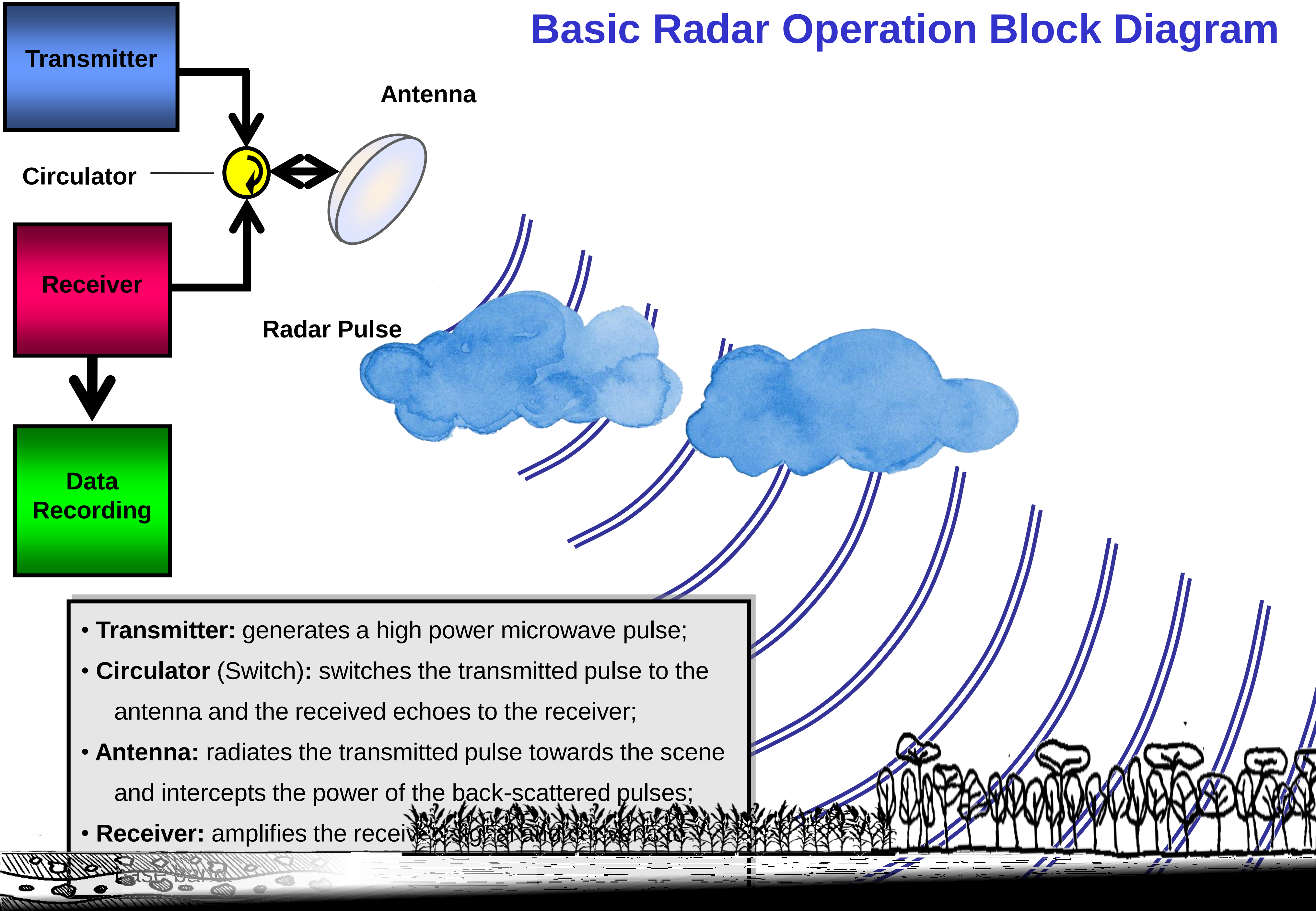
- **Transmitter:** generates a high power microwave pulse;
- **Circulator** (Switch)**:** switches the transmitted pulse to the antenna and the received echoes to the receiver;
- **Antenna:** radiates the transmitted pulse towards the scene and intercepts the power of the back-scattered pulses;
- **Receiver:** amplifies the received signal and converts to base band.

Basic Radar Operation Block Diagram



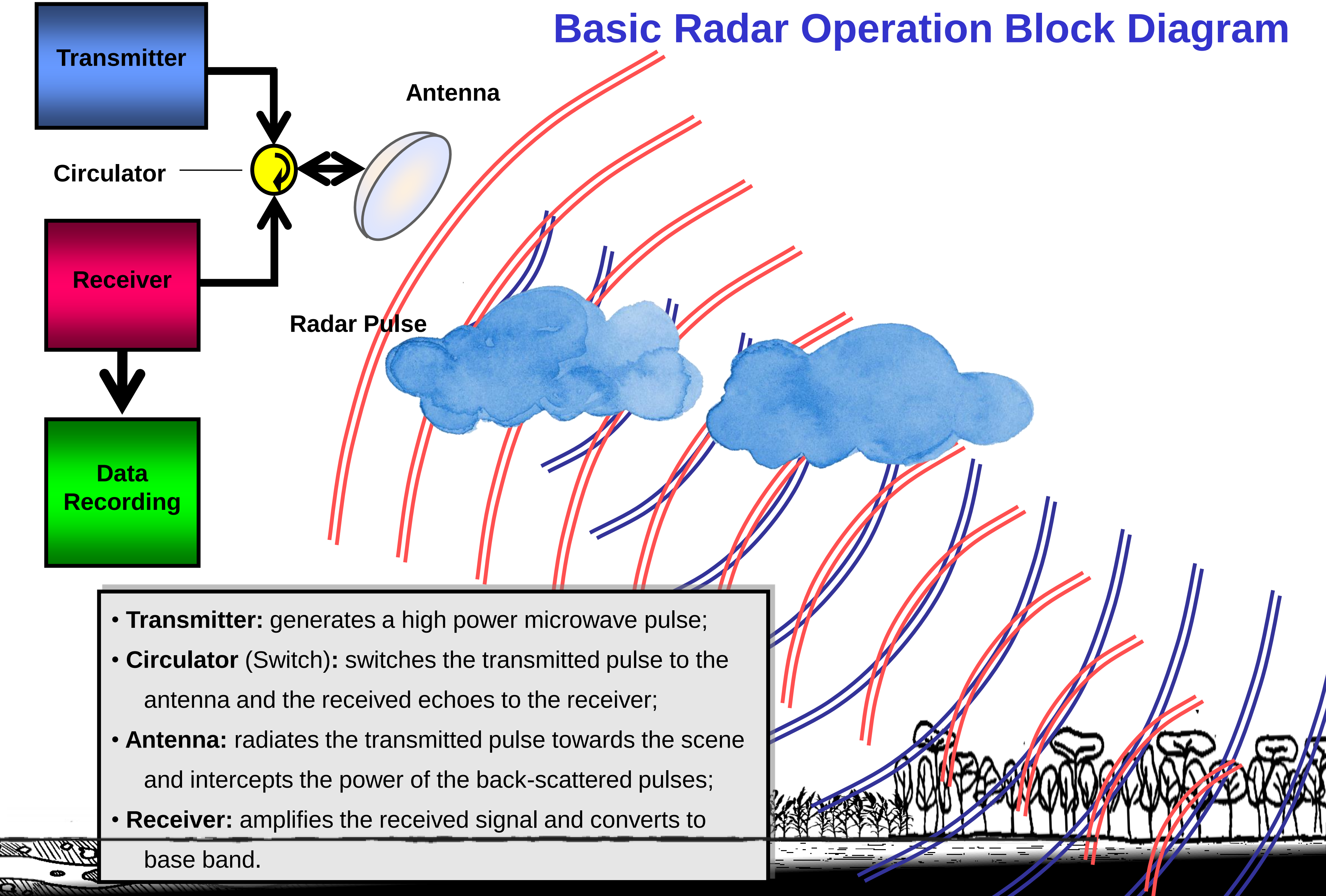
- **Transmitter:** generates a high power microwave pulse;
- **Circulator** (Switch)**:** switches the transmitted pulse to the antenna and the received echoes to the receiver;
- **Antenna:** radiates the transmitted pulse towards the scene and intercepts the power of the back-scattered pulses;
- **Receiver:** amplifies the received signal and converts to base band.

Basic Radar Operation Block Diagram



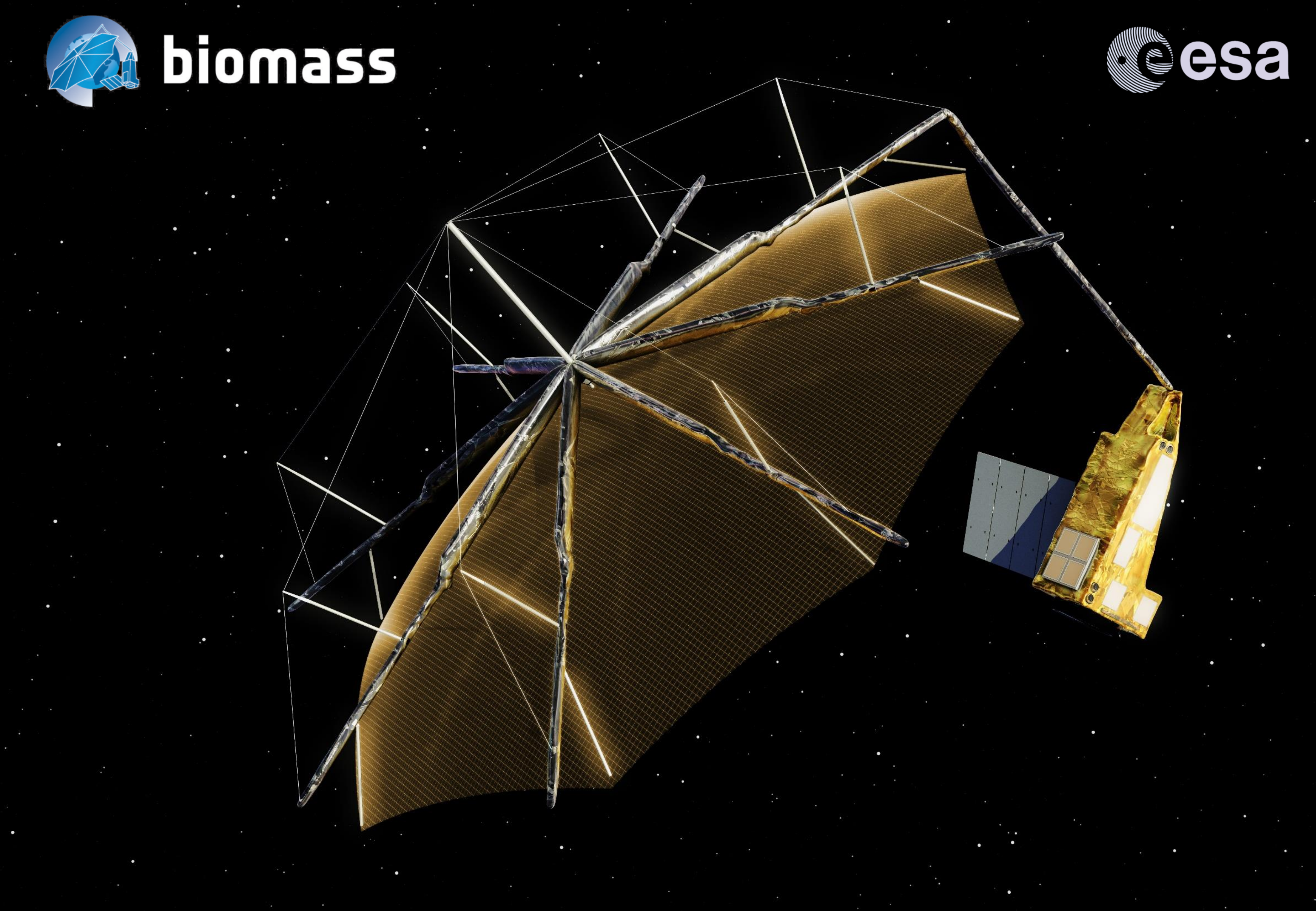
- **Transmitter:** generates a high power microwave pulse;
- **Circulator** (Switch)**:** switches the transmitted pulse to the antenna and the received echoes to the receiver;
- **Antenna:** radiates the transmitted pulse towards the scene and intercepts the power of the back-scattered pulses;
- **Receiver:** amplifies the receiv[illegible]

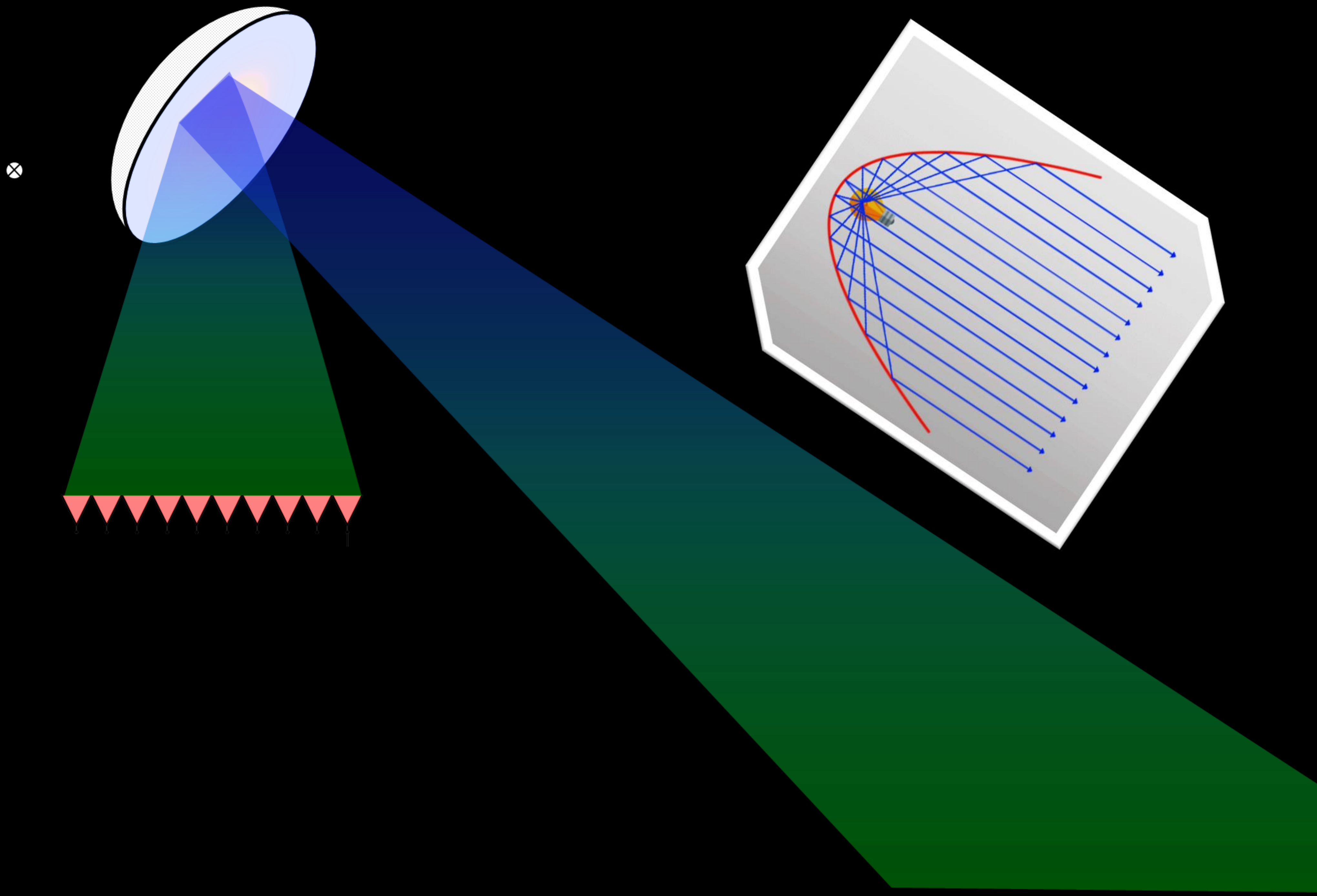
Basic Radar Operation Block Diagram



- **Transmitter:** generates a high power microwave pulse;
- **Circulator** (Switch)**:** switches the transmitted pulse to the antenna and the received echoes to the receiver;
- **Antenna:** radiates the transmitted pulse towards the scene and intercepts the power of the back-scattered pulses;
- **Receiver:** amplifies the received signal and converts to base band.

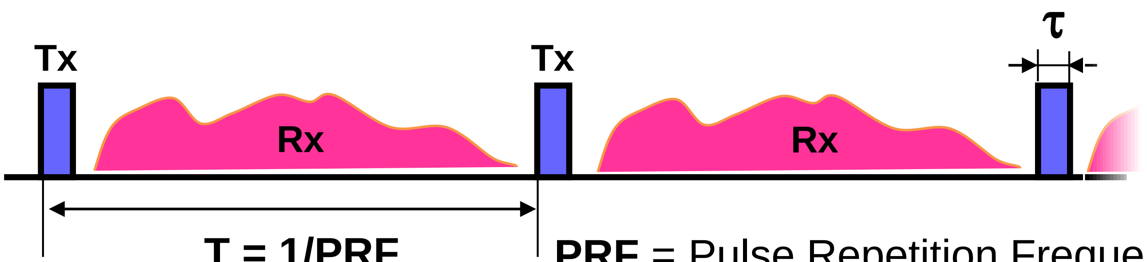

sentinel-1
esa


biomass
esa

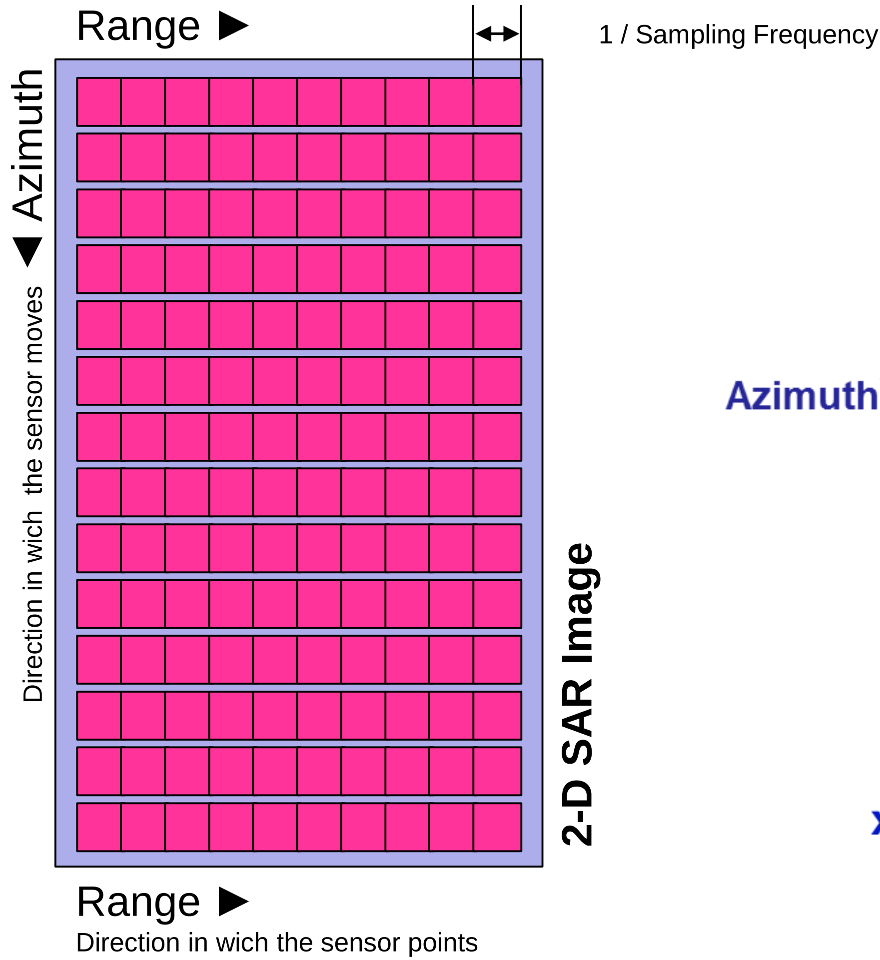


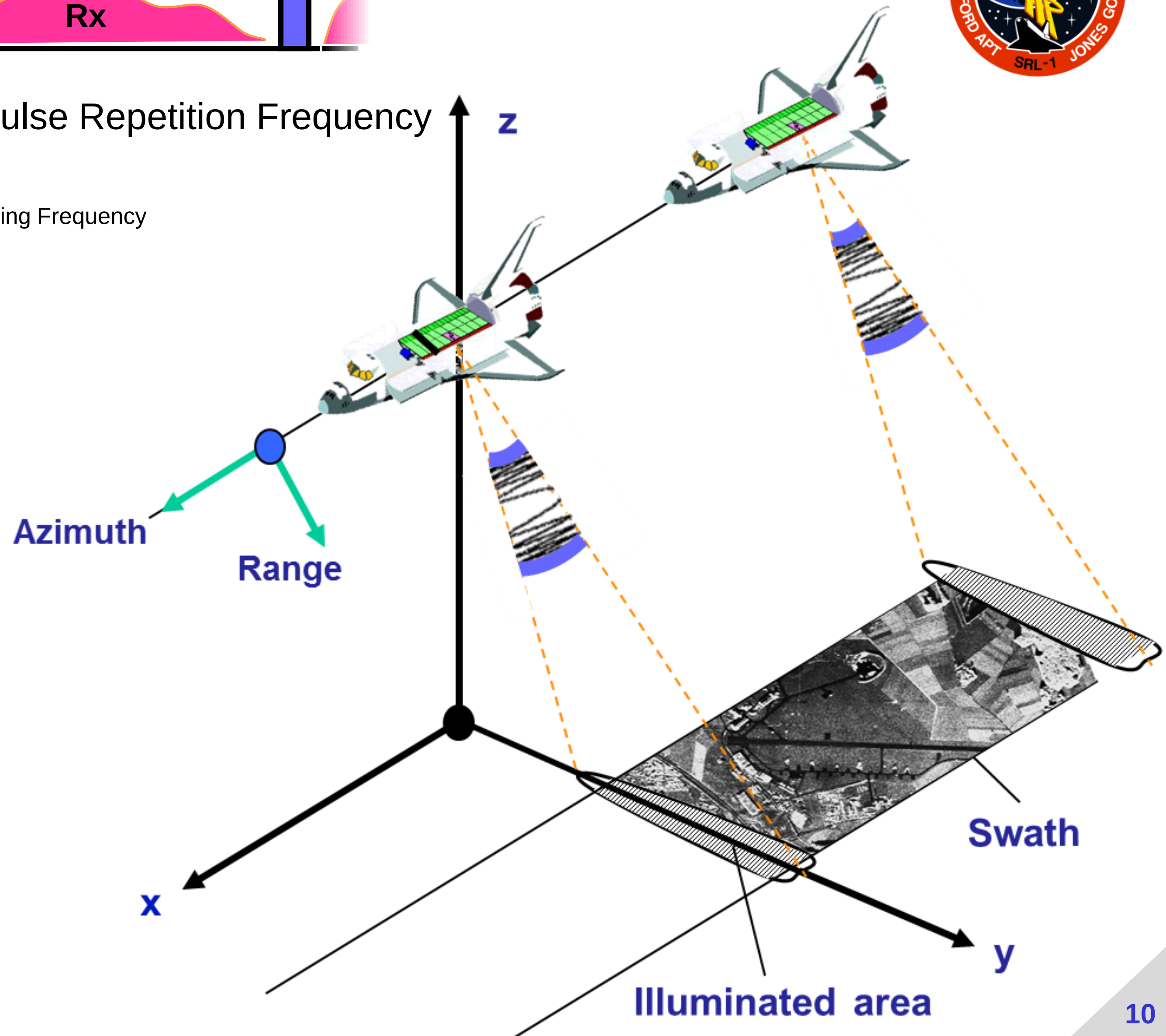
2-D Imaging

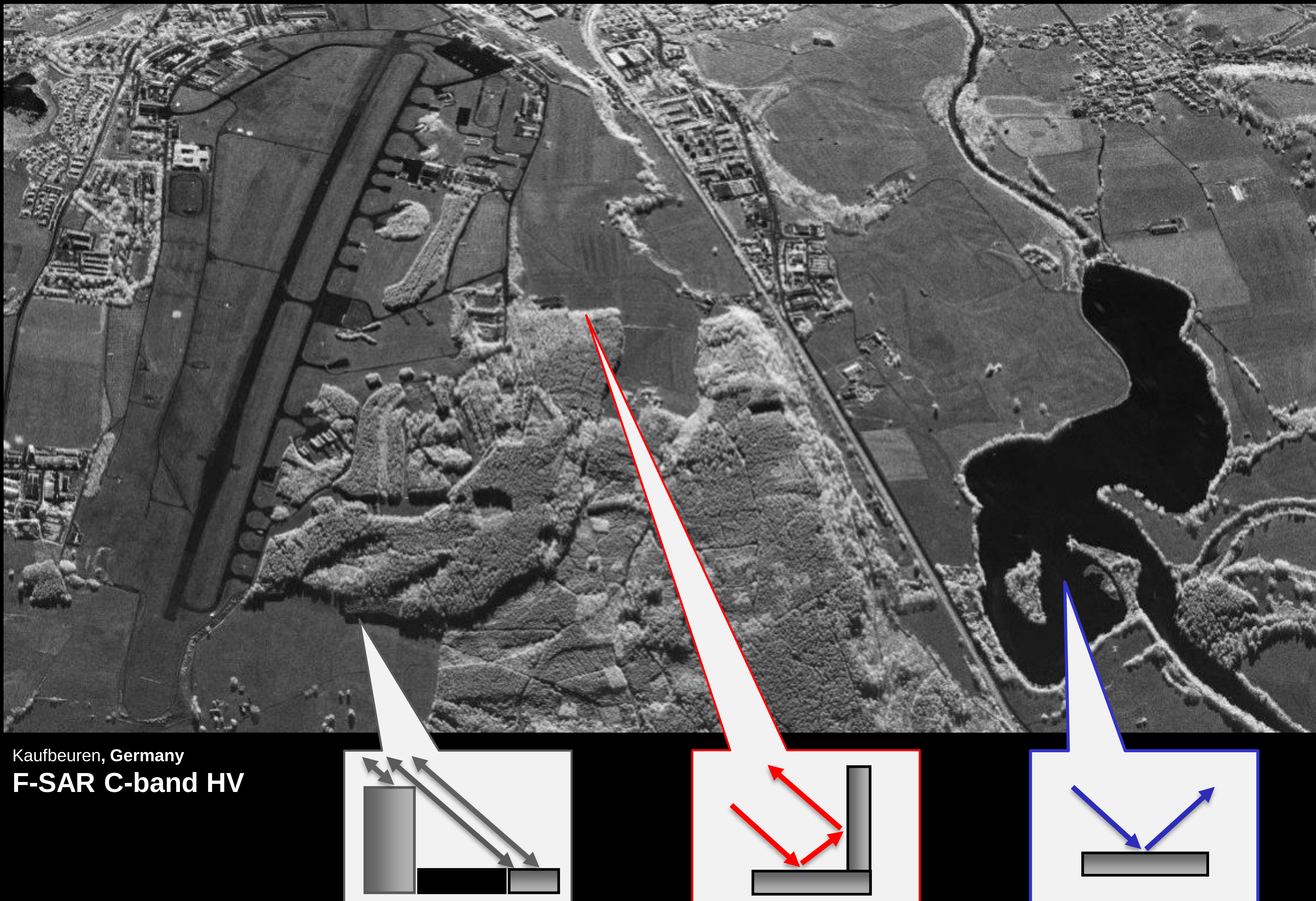
Pulsed radar on a moving platform



PRF = Pulse Repetition Frequency



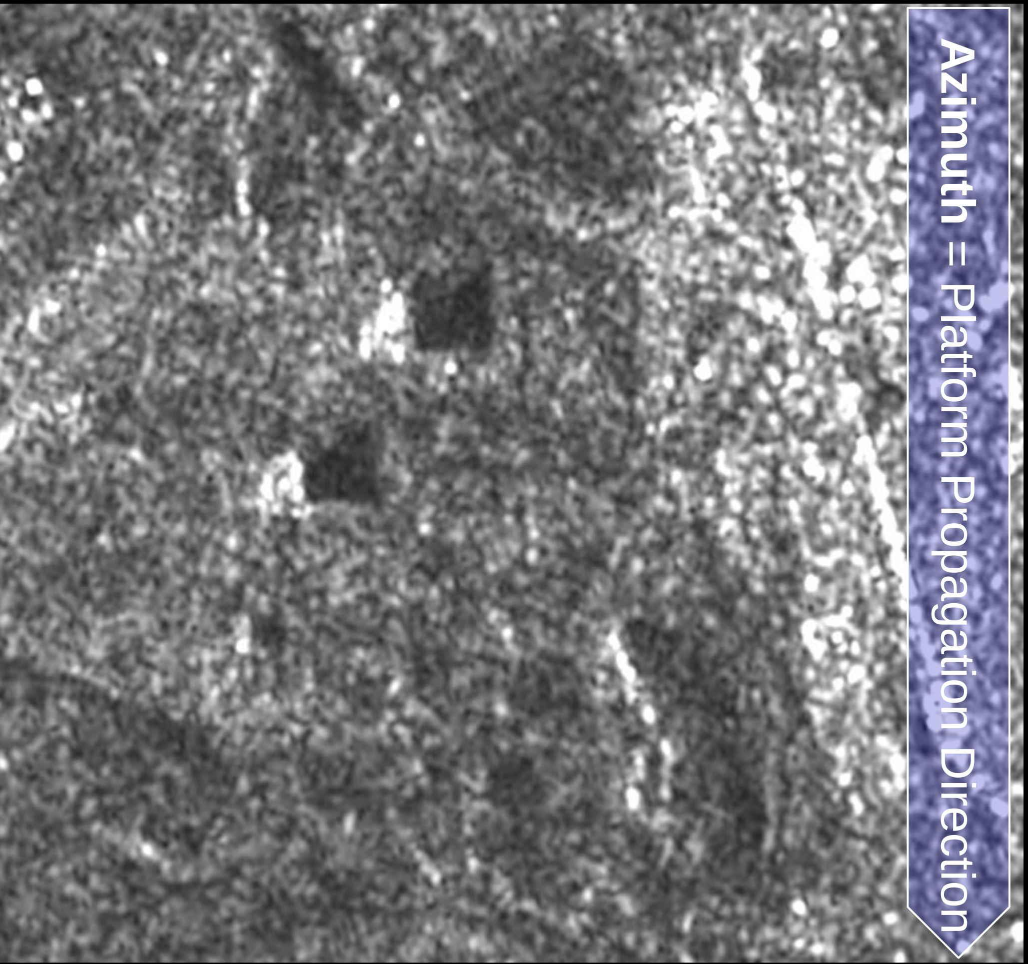




Kaufbeuren, **Germany**

F-SAR C-band HV

Spatial Resolution

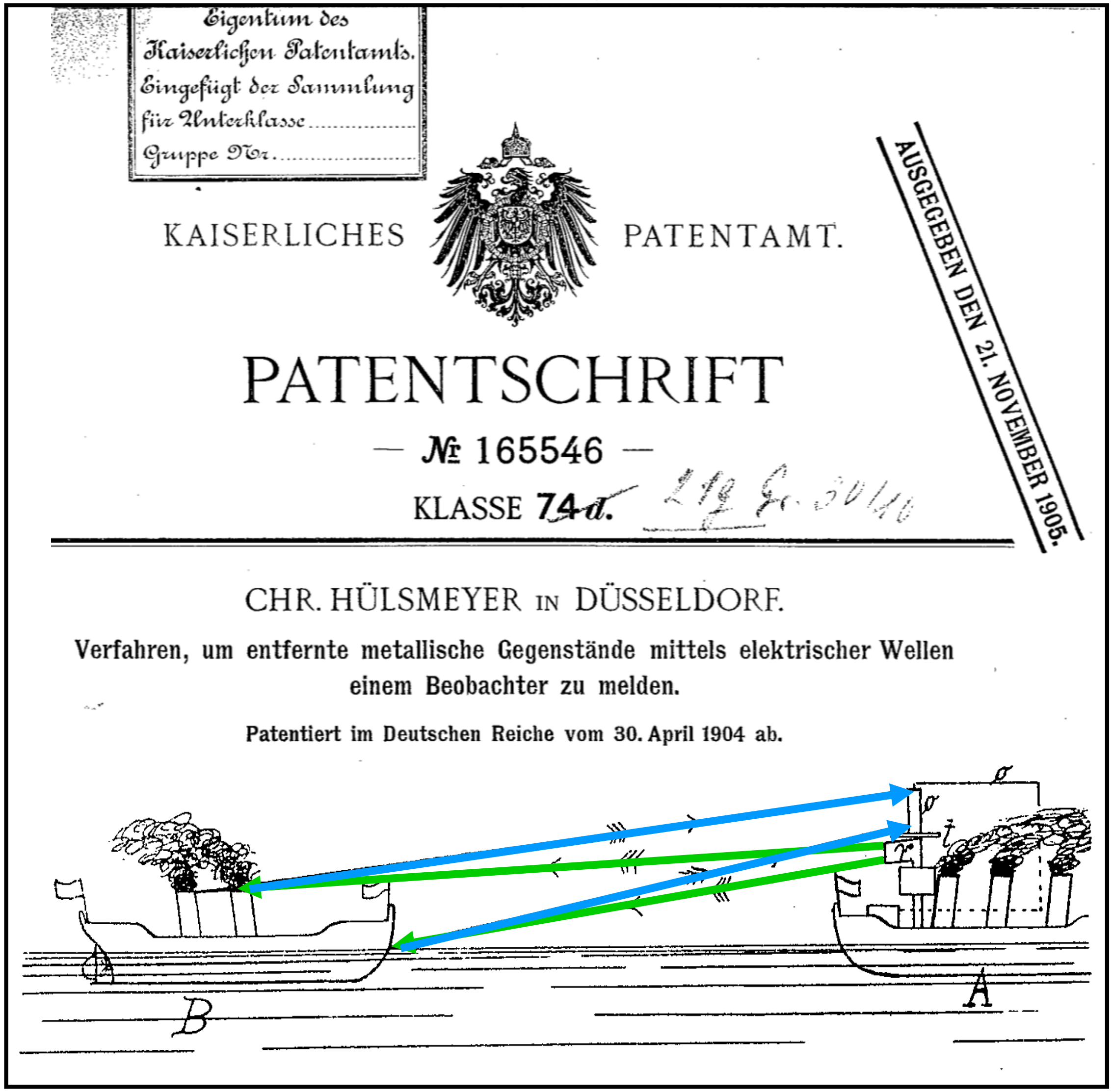


20m Resolution (X-SAR - 2000)

1m Resolution (TerraSAR-X)


Range Resolution

Christian Hülsmeyer: Inventor of Radar (1904)


Eigentum des Kaiserlichen Patentamts.
Eingefügt der Sammlung
für Unterklasse ....
Gruppe Nr. ....

KAISERLICHES PATENTAMT.

AUSGEGEBEN DEN 21. NOVEMBER 1905.

PATENTSCHRIFT

— № 165546 —

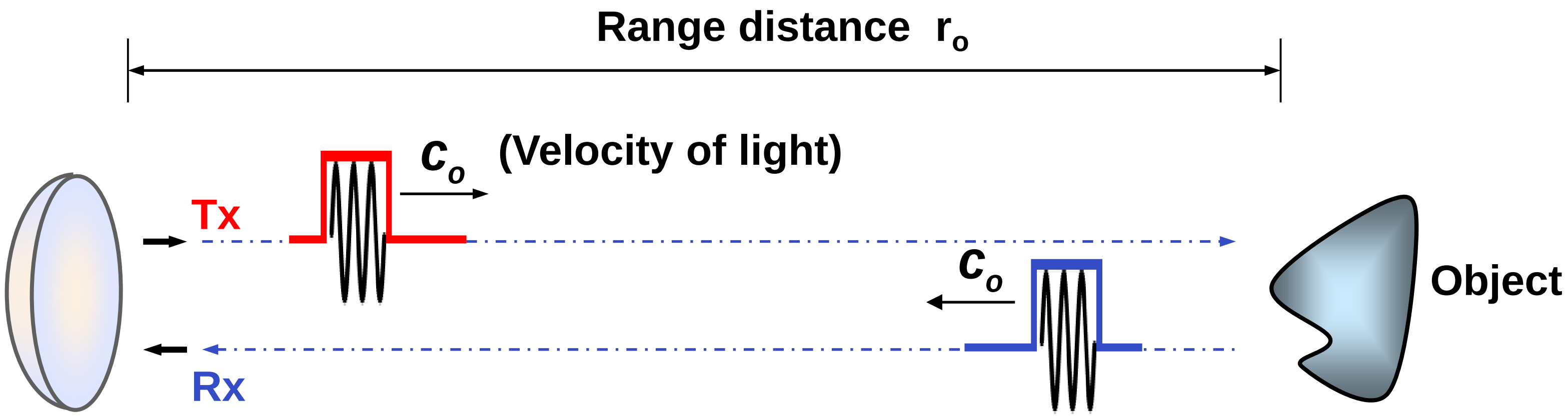
KLASSE 74d.

CHR. HÜLSMEYER IN DÜSSELDORF.

Verfahren, um entfernte metallische Gegenstände mittels elektrischer Wellen einem Beobachter zu melden.

Patentiert im Deutschen Reiche vom 30. April 1904 ab.

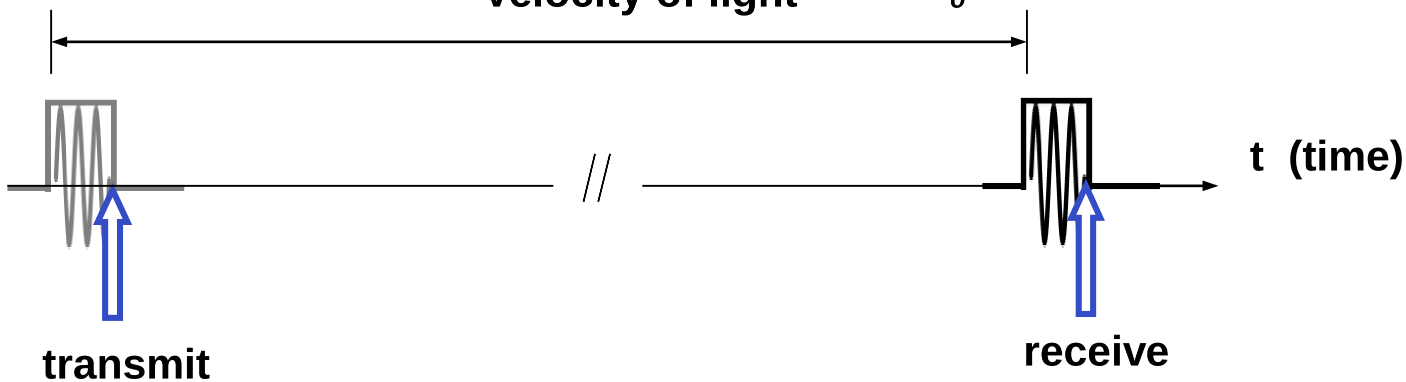
Radar Measurement Principle



Received echo signal (back-scattered signal of imaged object):

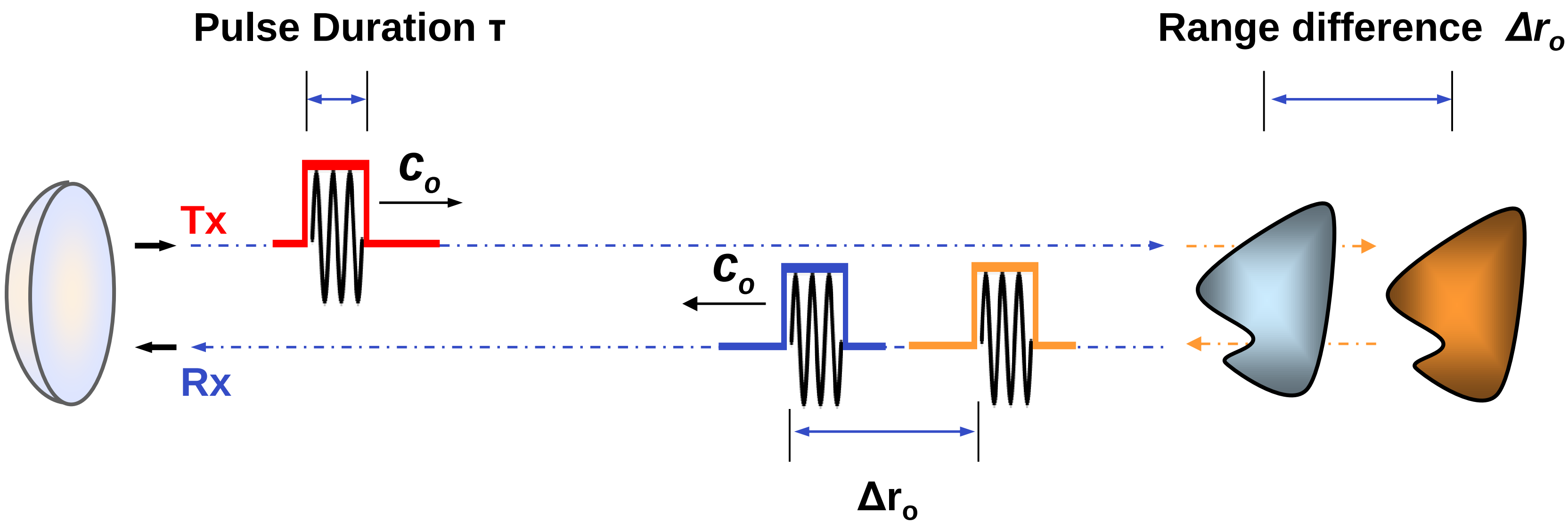
$$\text{Total time delay } t = \frac{\text{Total distance}}{\text{Velocity of light}} = \frac{2 \cdot r_o}{c_o}$$

c_0 = 299 792 458 m/s



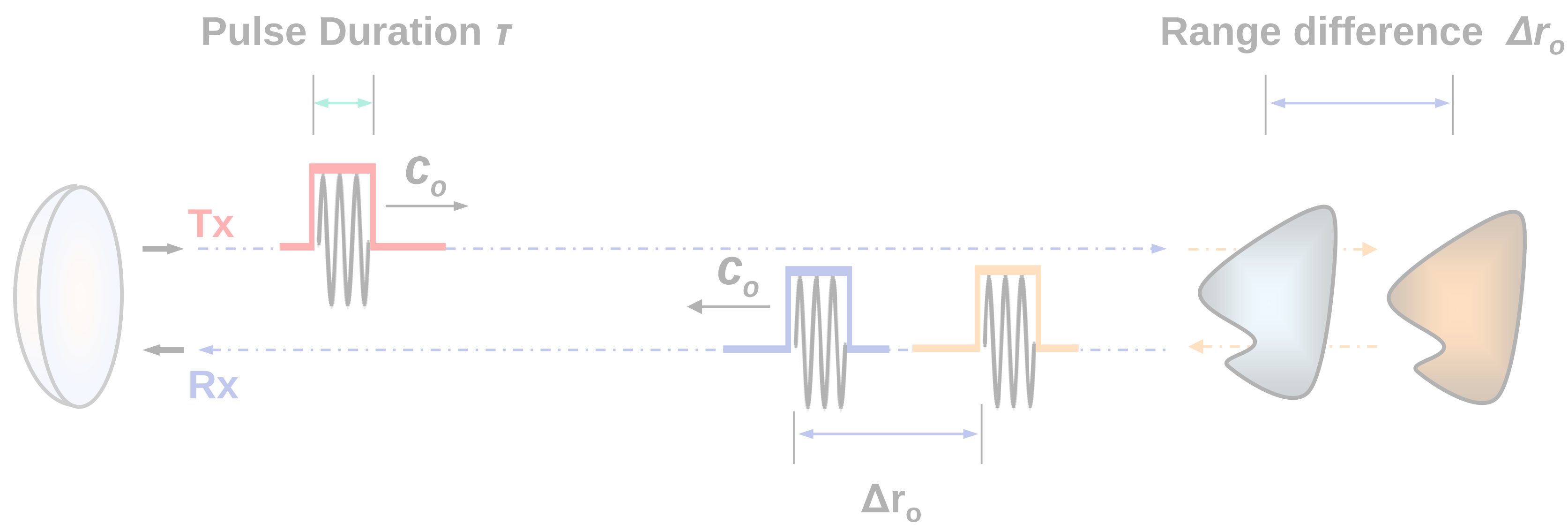
Example: A time delay of t=1µsec corresponds to a distance of r_0=150m.

(Slant) Range Resolution

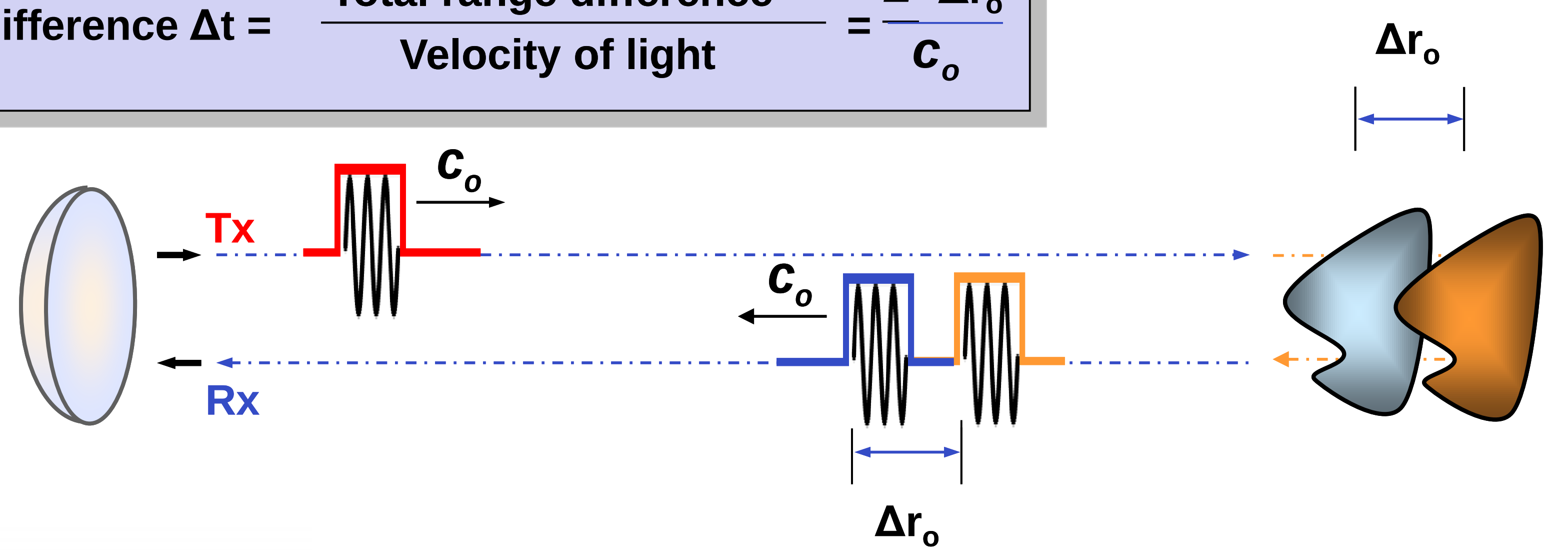


$$\text{Time difference } \Delta t = \frac{\text{Total range difference}}{\text{Velocity of light}} = \frac{2\,\Delta r_o}{c_o}$$

(Slant) Range Resolution

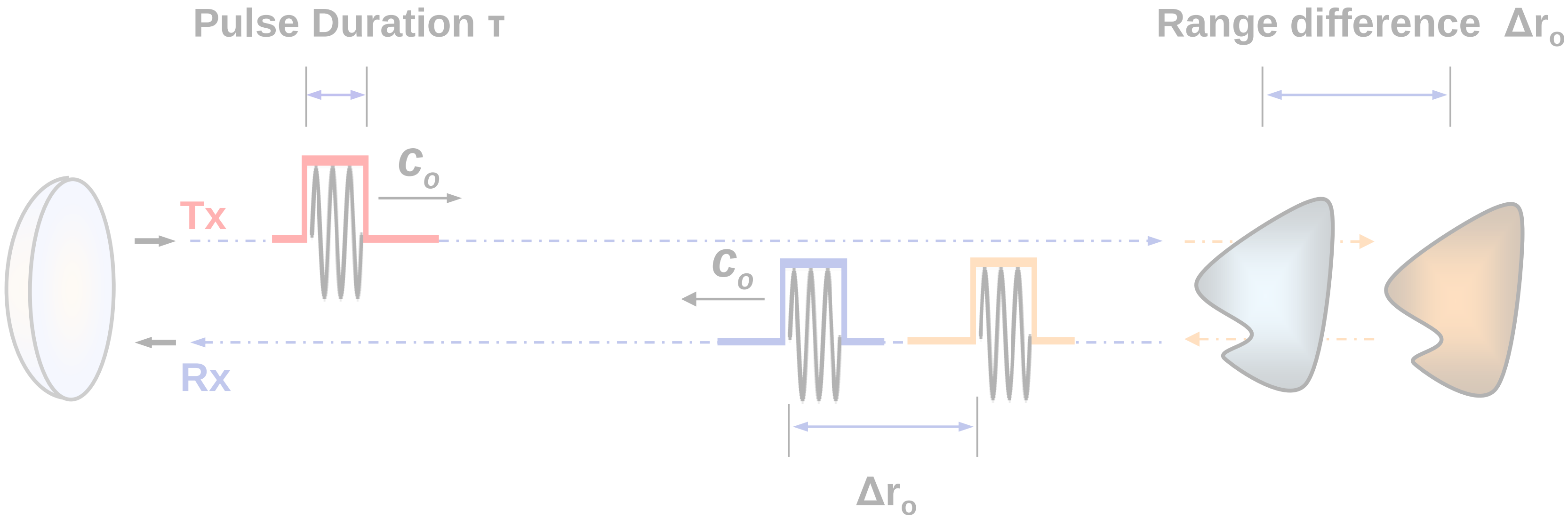


$$\text{Time difference } \Delta t = \frac{\text{Total range difference}}{\text{Velocity of light}} = \frac{2\ \Delta r_o}{c_o}$$

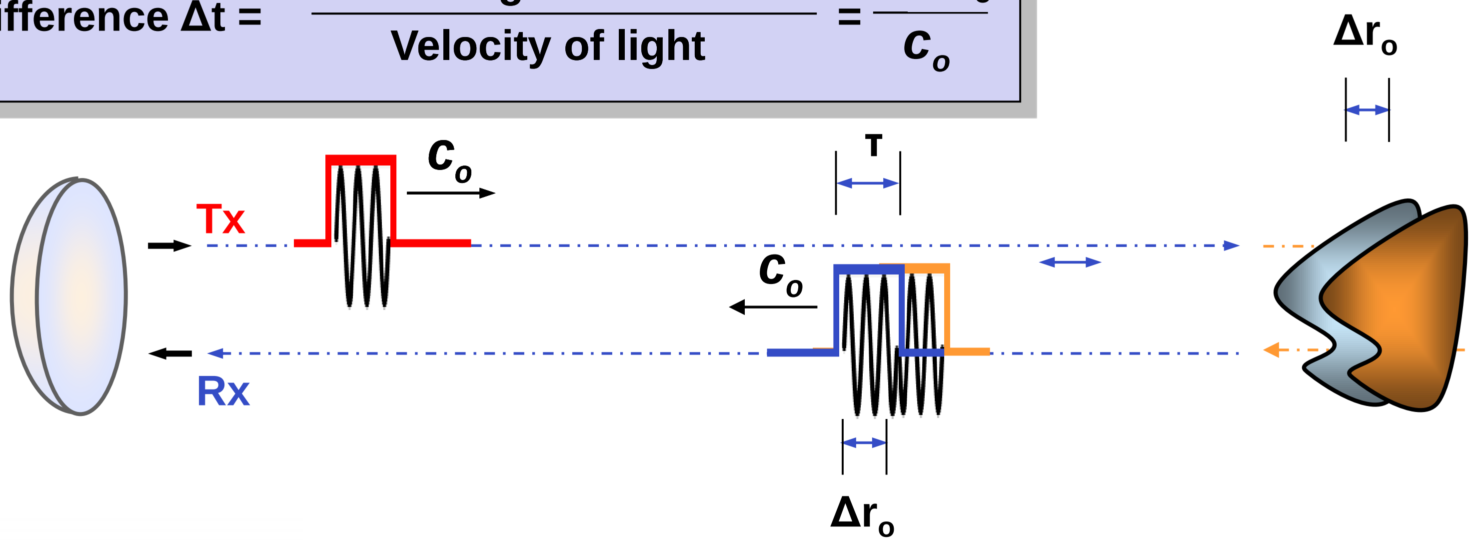


Two scatterers are resolved because they have different time delays; i.e. because their echoes arrive at different times.

(Slant) Range Resolution

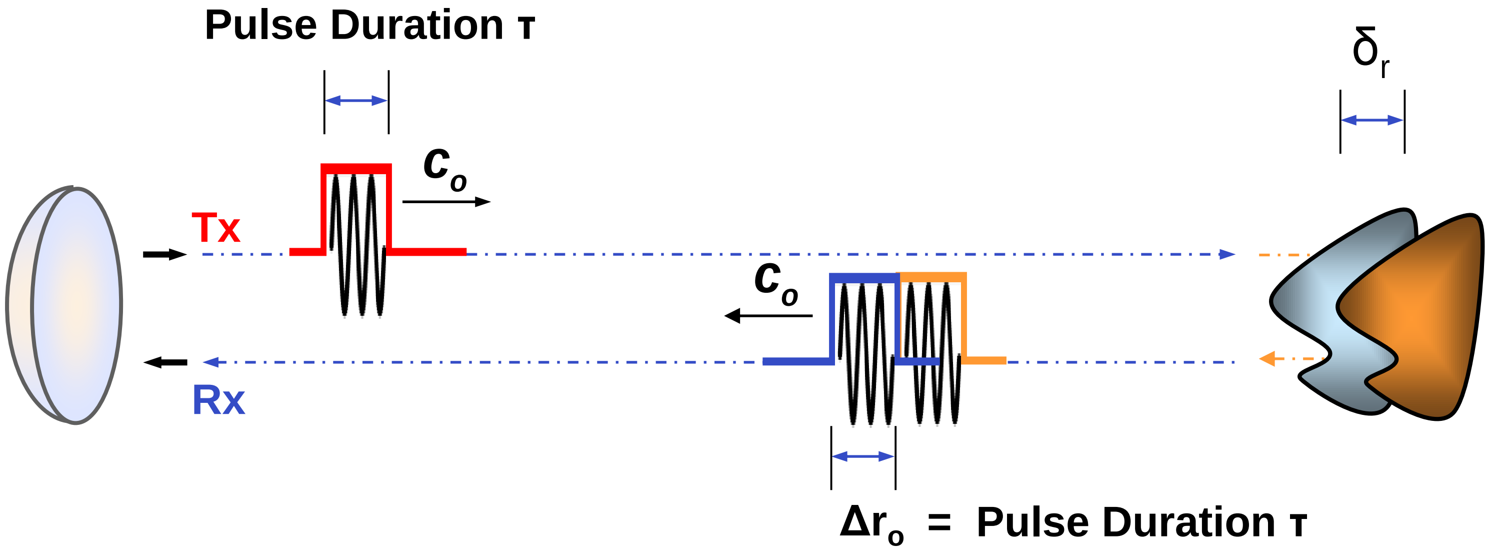


$$\text{Time difference } \Delta t = \frac{\text{Total range difference}}{\text{Velocity of light}} = \frac{2\ \Delta r_o}{c_o}$$



When the time difference Δt becomes smaller than the pulse duration τ then the objects cannot be separated from each other !

(Slant) Range Resolution



Range resolution: The min. distance between two objects that can still be separated from each other i.e. the distance that corresponds to a time difference Δt equal than the pulse duration τ

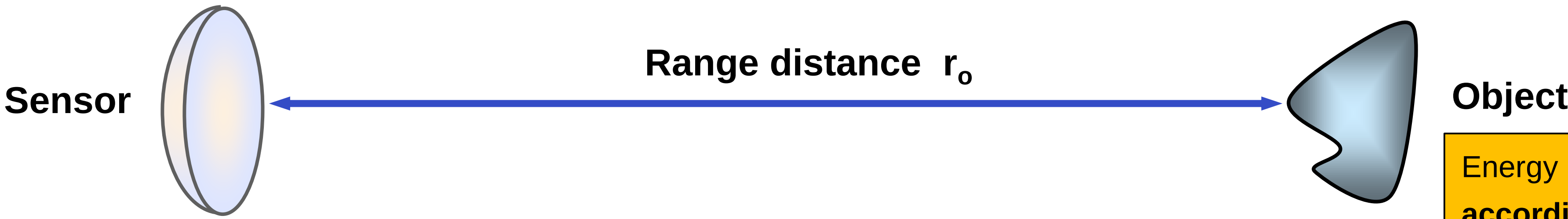
Time difference Δt := Pulse duration τ = $\dfrac{\textbf{2 Range resolution}}{\textbf{Velocity of light}} = \dfrac{2\delta_r}{c_0} \rightarrow$ **Rg. resolution:** $\delta_r = \dfrac{c_0 \tau}{2}$

The range resolution is independent on the distance between the sensor and the scatterer; it only depends on the pulse duration τ that is inverse proportional to the pulse bandwidth W: $\delta_r = \dfrac{c_0 \tau}{2} = \dfrac{c_0}{2W}$

Example: A pulse bandwidth of W=100MHz leads to a range resolution δ_r=1.5m (corresponding to a pulse duration of τ=10ns) .

The Dilemma

The **energy of the transmitted pulse** defines the (max.) distance an object can be sensed:



Received signal energy E_{RX} from an object at distance r_0: $\boxed{E_{Rx} \sim \frac{E_{Tx}}{r_0^4}}$

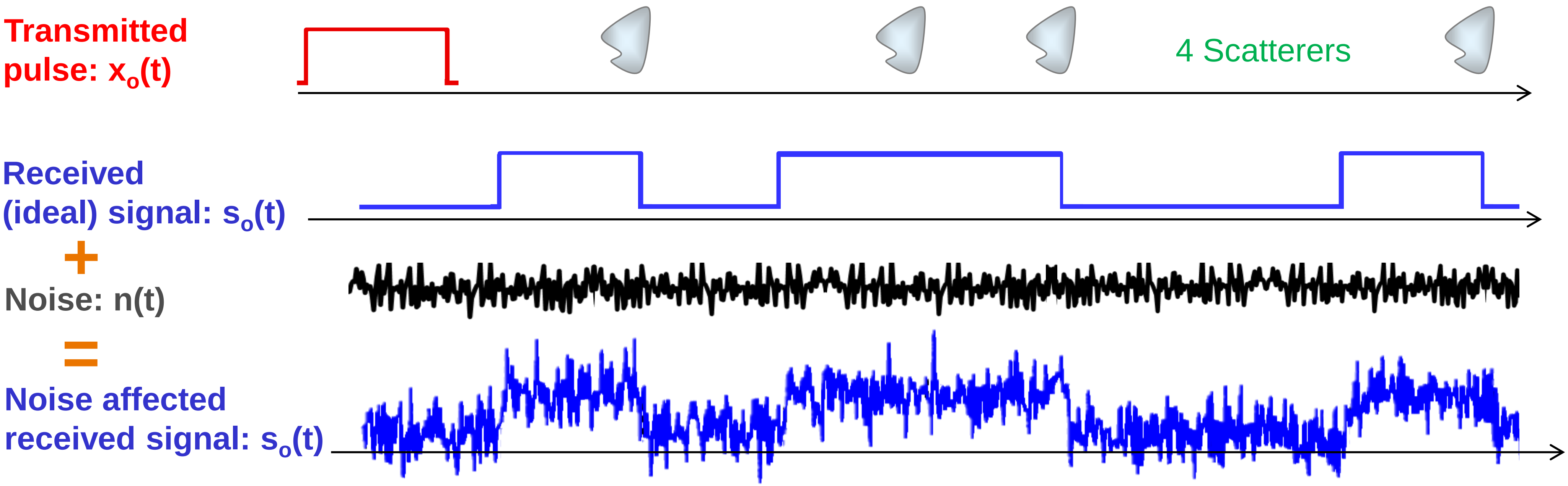
Energy decay's **according to the inverse 4th power of the distance** to the object !!!

- The **energy of the transmitted pulse** E_{Tx} is given by the product of instantaneous peak power **P** and pulse duration τ: $E_{Tx} = P_{Tx}\tau$
- **P** is limited by the sensor hardware – especially in the case of spaceborne sensors so that for **increasing the pulse energy long pulses are required**.

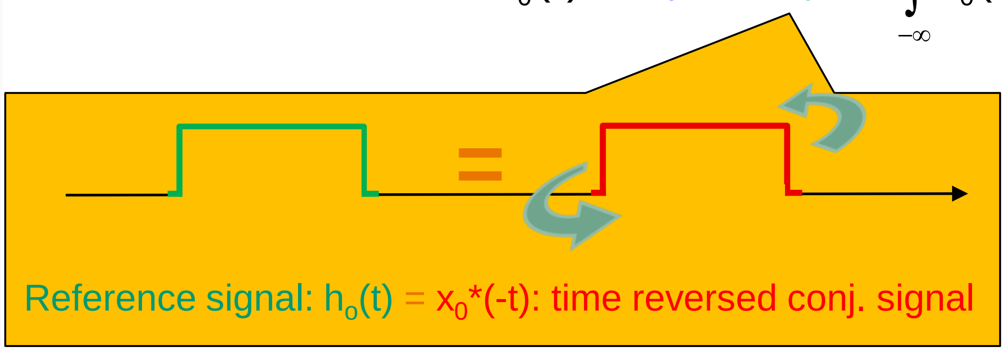
Limitation: For **high spatial resolution short pulses** (realized by wide bandwidths) are required.

The solution: To employ (frequency) **modulated pulses** (FM Chirp) and **matching filtering** !

The Matched Filter



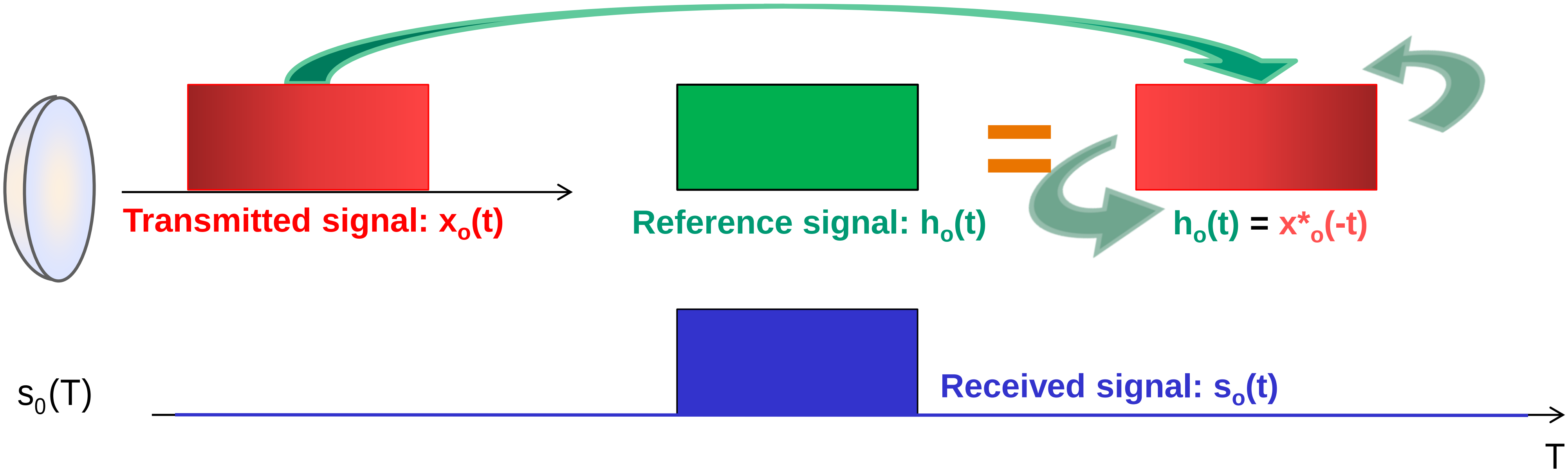
Matched Filter: $u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T)\cdot h_0(t-T)\; dT$ where $\otimes$ stands for convolution



Matched filters allow the optimal detection of a known signal template (e.g. the transmitted pulse) in an noise-affected signal sequence (e.g. received signal).
This is achieved by convolving the signal sequence with a conjugate time-reversed copy of the signal template.

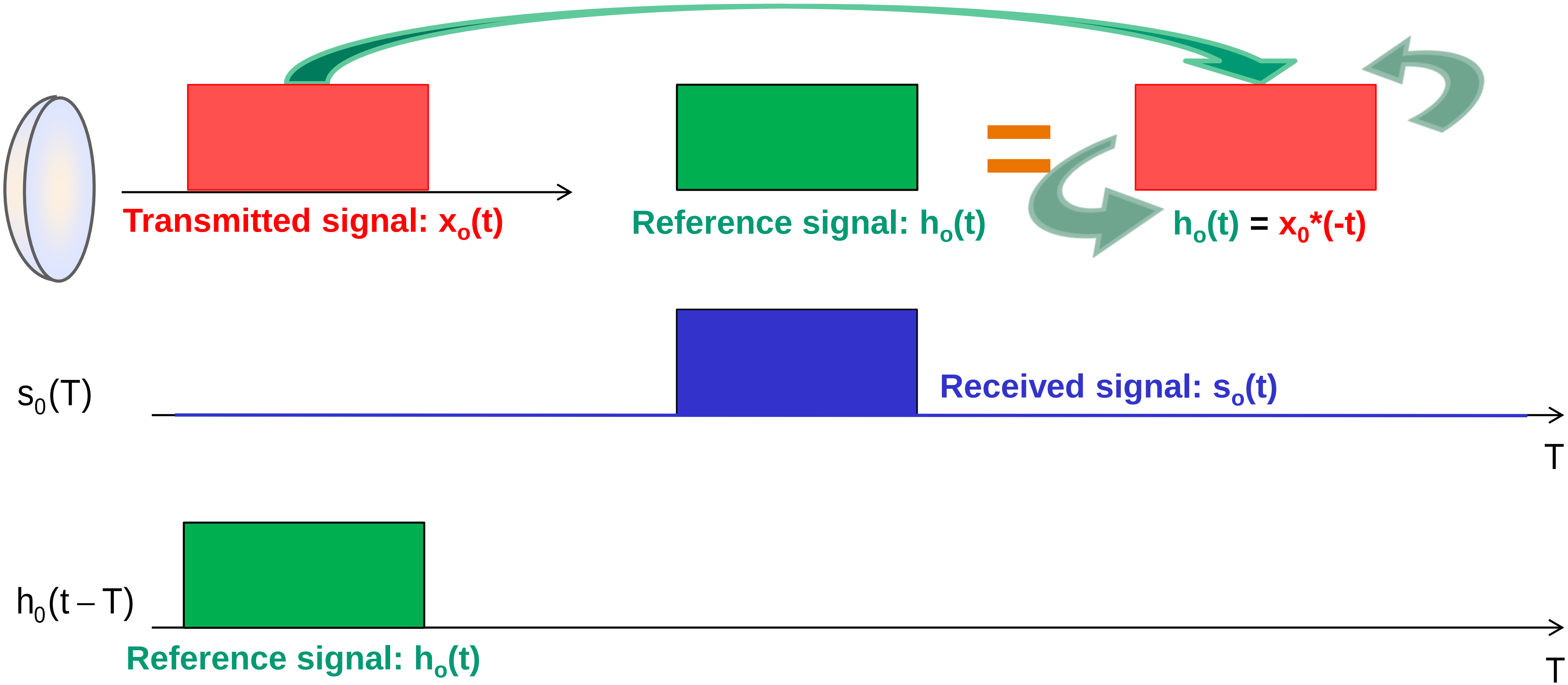
Matched Filter

$u_0(t) = s_0(t) \otimes h_0(t)$



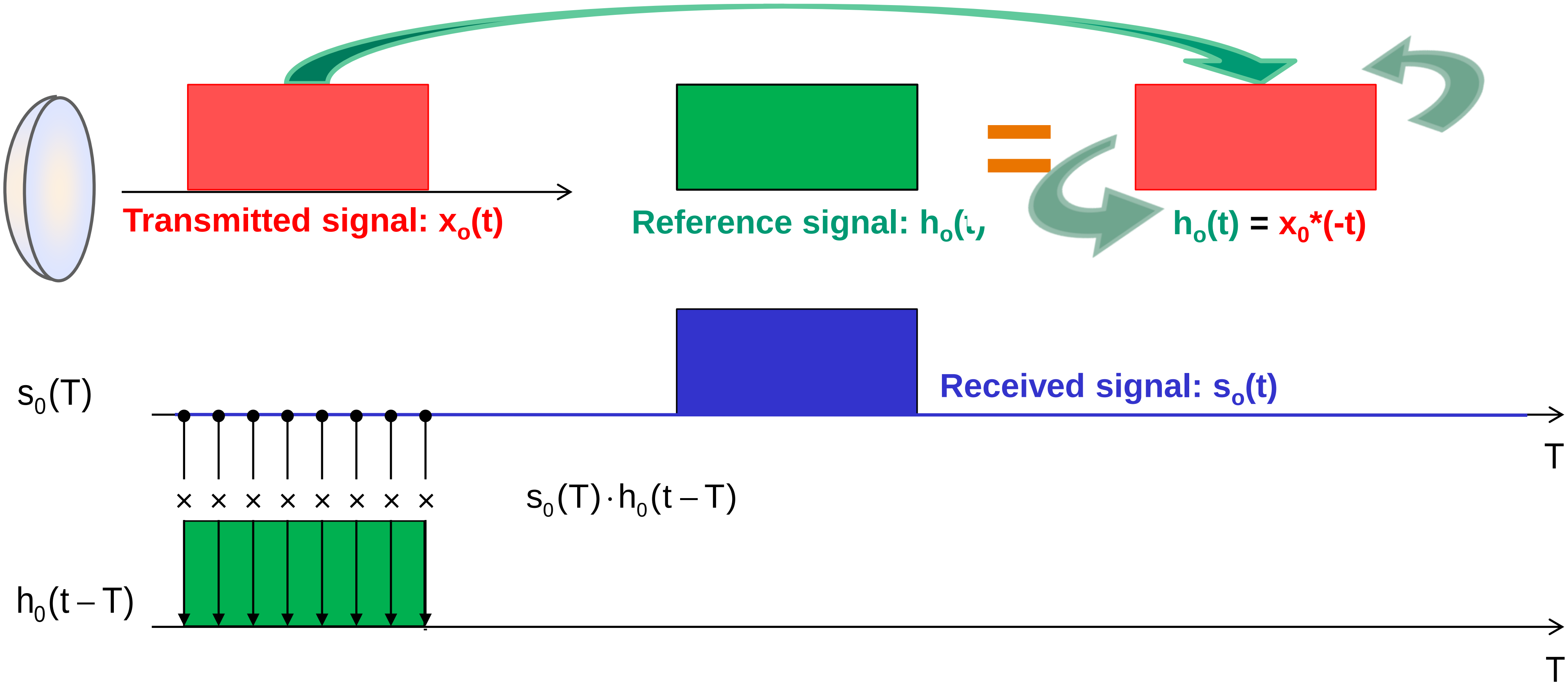
Matched Filter

$$u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t-T)\ dT = \int_{-\infty}^{\infty} s_0(T) \cdot x_0^*(T-t)\ dT$$



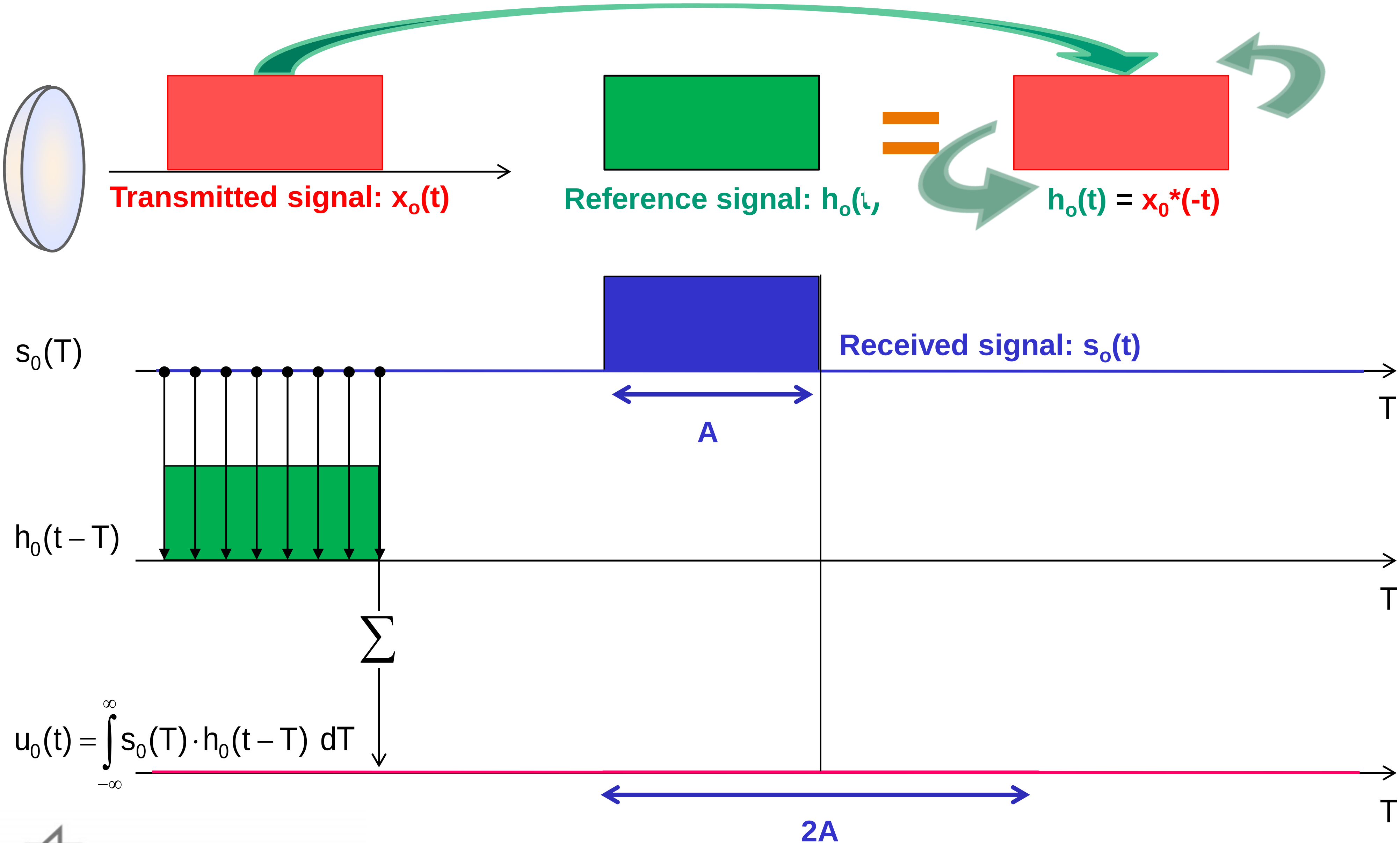
Matched Filter

$$u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T)\cdot h_0(t-T)\ dT = \int_{-\infty}^{\infty} s_0(T)\cdot x_0^*(T-t)\ dT$$

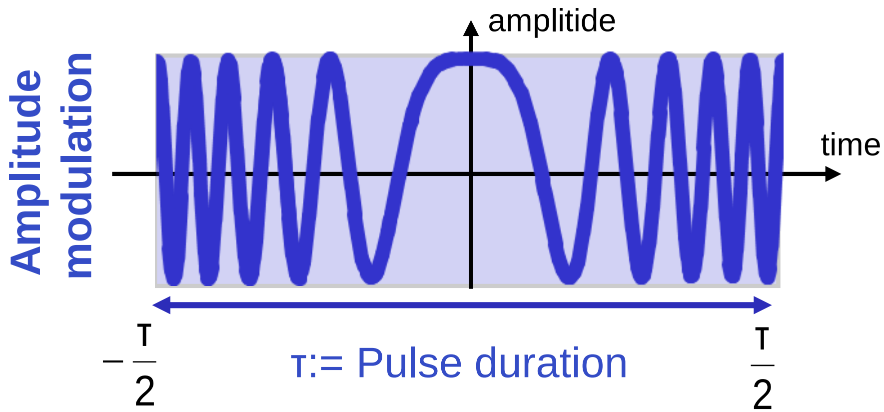


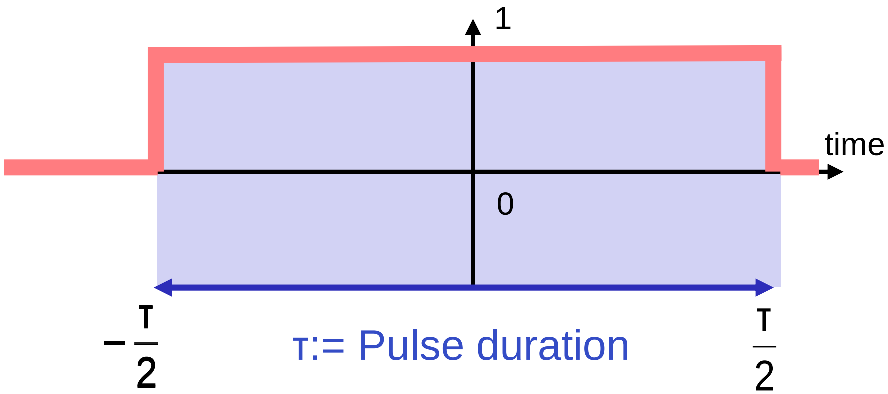
Matched Filter

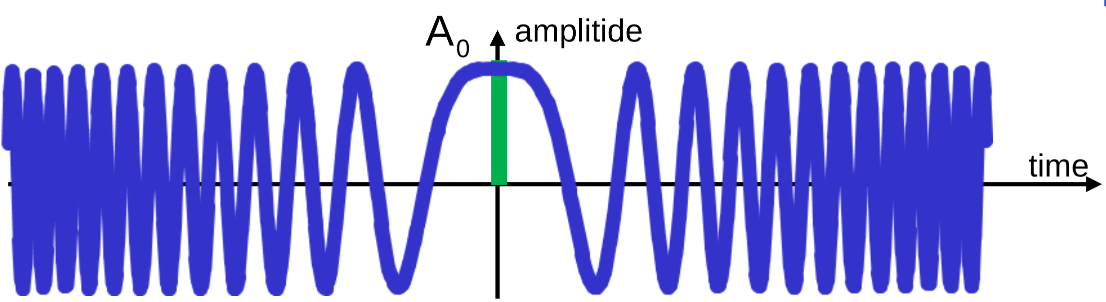
$$u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t-T)\ dT = \int_{-\infty}^{\infty} s_0(T) \cdot x_0^*(T-t)\ dT$$



Linear Frequency Modulated Pulse: FM Chirp







Amplitude

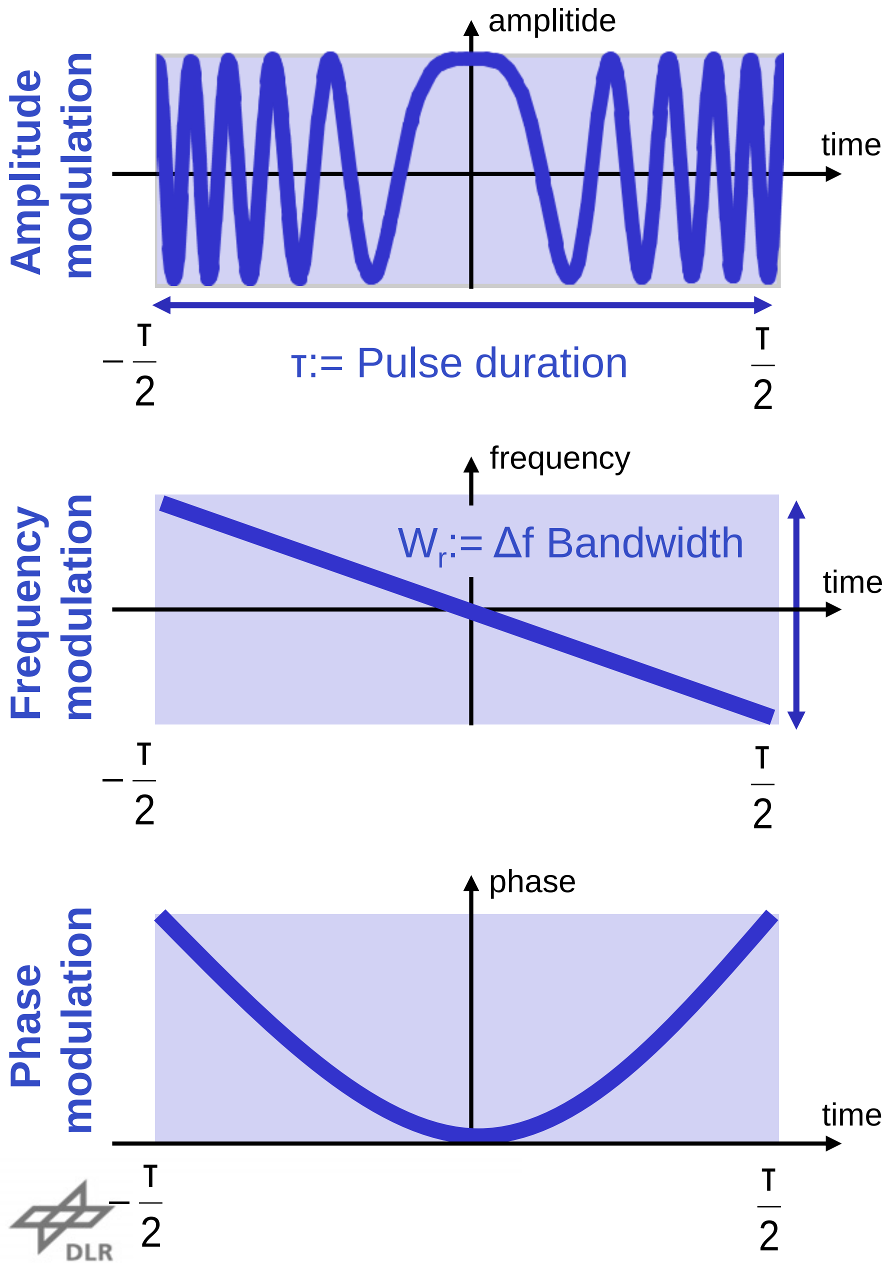
(Frequency) Modulation

Window

Chirp: $x_0(t) = \underbrace{A_0}_{\text{Amplitude}}\ \underbrace{e^{-i\varphi_r(t)}}_{\text{(Frequency) Modulation}}\ \underbrace{\mathrm{rect}\left(\frac{t}{T}\right)}_{\text{Window}} = A_0\, e^{-i\pi k_r t^2}\, \mathrm{rect}\left(\frac{t}{T}\right)$

Frequency variation: $f_r(t) = k_r t$ with $-\frac{T}{2} \le t \le \frac{T}{2}$

Linear Frequency Modulated Pulse: FM Chirp



FM Chirp: A pulse in which the frequency f_r is linearly increasing (or degreasing) with time (along the pulse)

Frequency variation: $f_r(t) = k_r t$ with $-\frac{\tau}{2} \le t \le \frac{\tau}{2}$

k_r may be positive (Up-) or negative (Down-Chirp)

The pulse bandwidth $W_r := \Delta f$ does no longer depend on the pulse length τ. So it becomes possible to generate pulses with large pulse lengths τ and large bandwidths W_r

Chirp: $x_0(t) = A_0\, e^{-i\varphi_r(t)}\, \mathrm{rect}\left(\frac{t}{\tau}\right) = A_0\, e^{-i\pi k_r t^2}\, \mathrm{rect}\left(\frac{t}{\tau}\right)$

Chirp bandwidth: $W_r := k_r \tau$

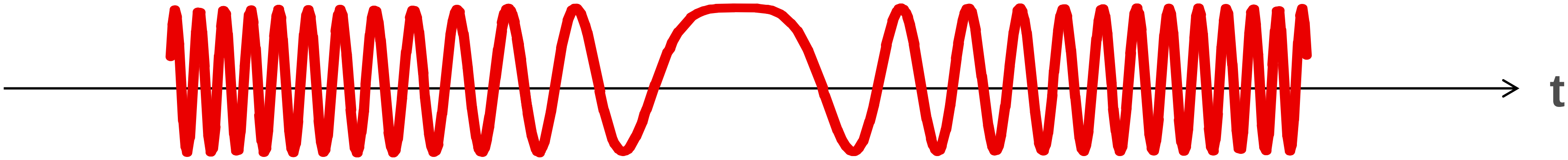
Pulse compression gain: $C_r = 10 \log_{10}(W_r \tau_r)$

Phase modulation: $\varphi_r(t) = 2\pi \int_{-\tau/2}^{+\tau/2} f_r(t')\, dt' = \pi\, k_r t^2 \Big|_{-\tau/2}^{+\tau/2}$

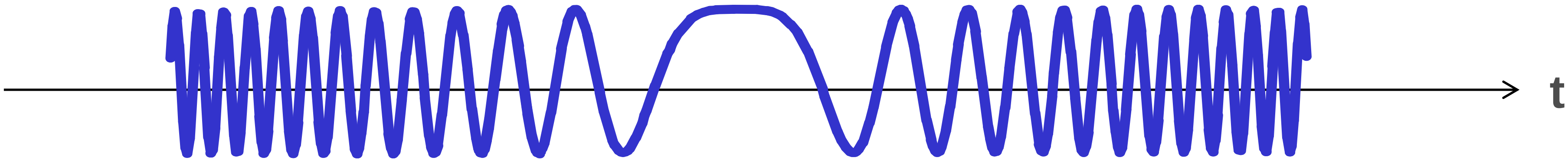
Note: The linear frequency modulation leads to a quadratic phase dependency on time.

Range Compression: Matched Filter

Transmitted Signal $x_o(t)$

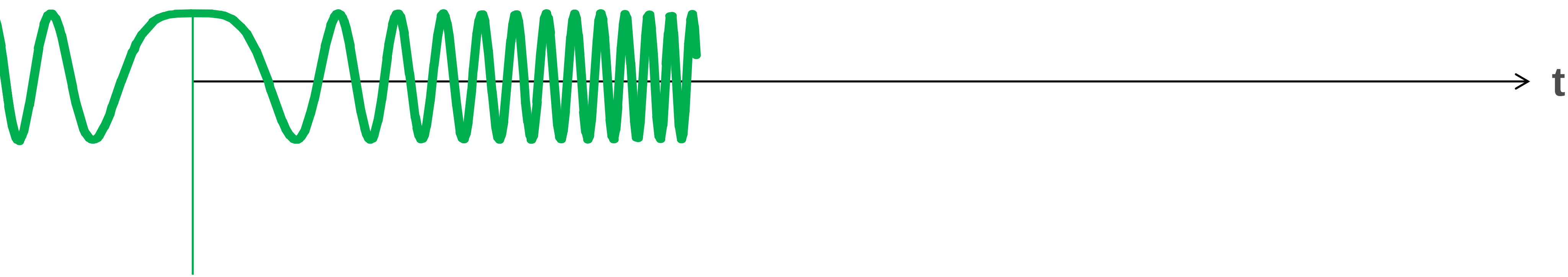


Received Signal $s_o(t)$



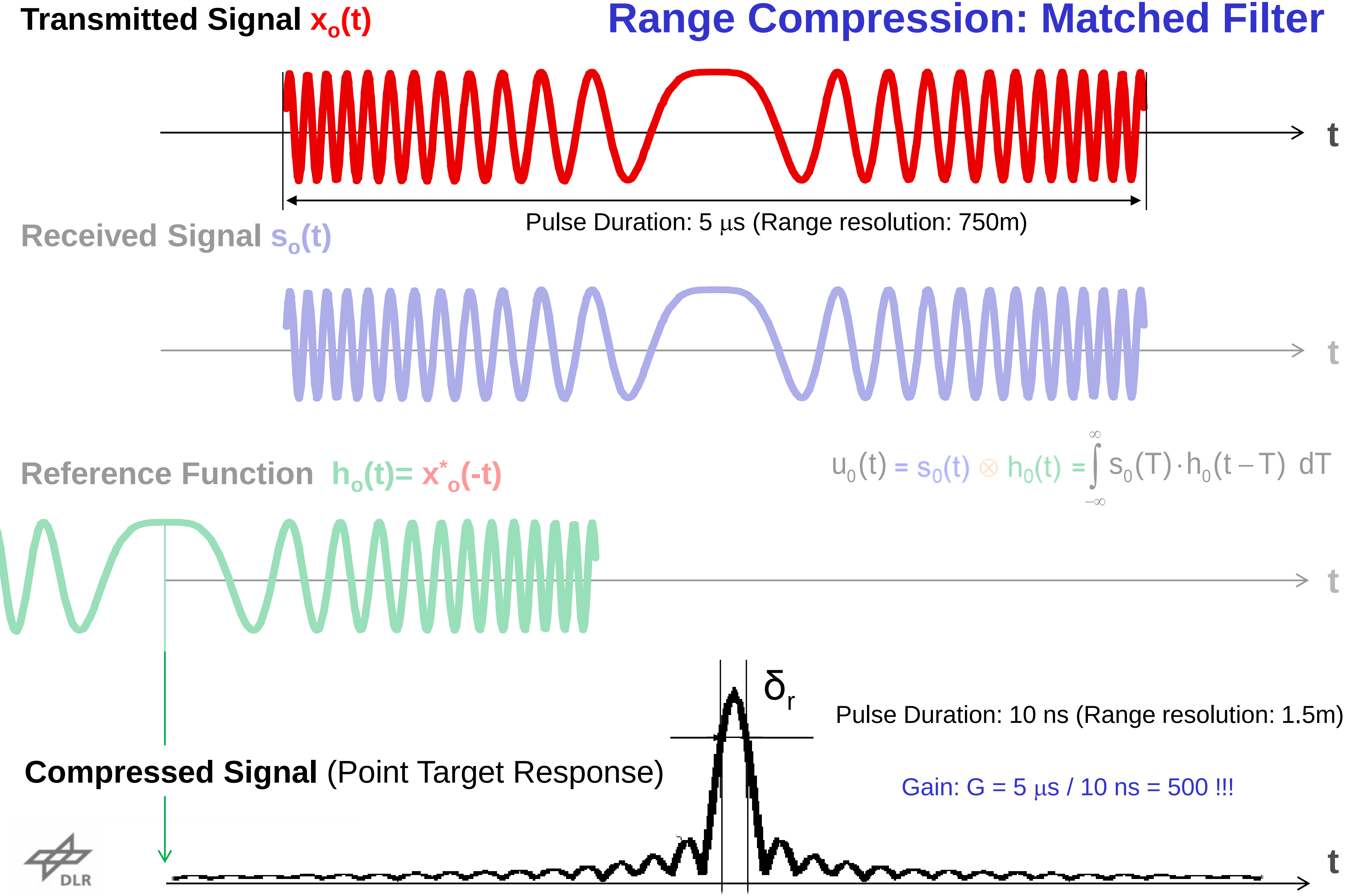
Reference Function $h_o(t) = x^*_o(-t)$

$$u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t - T)\ dT$$



Point Target Response

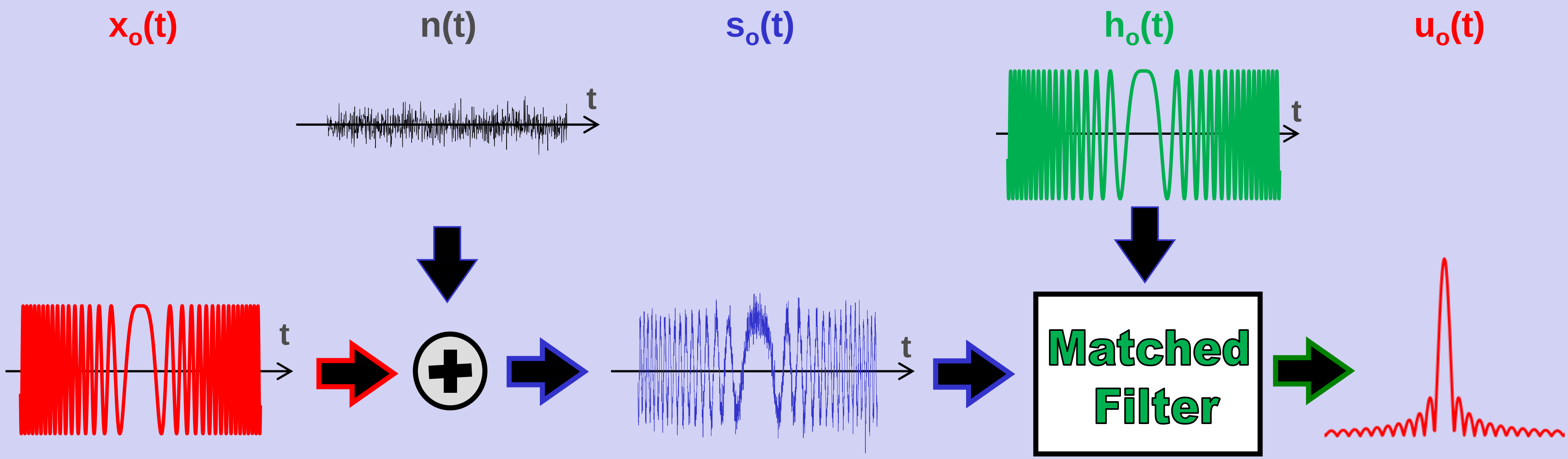
Range Compression: Matched Filter



Pulse Compression using Matched Filter

A long chirp pulse $x_0(t)$ is transmitted. Received is the transmitted pulse plus noise $s_0(t)=x_0(t)+n(t)$.

The high range resolution (e.g. the pulse compression) is achieved after reception: The long received pulse $s_o(t)$ is compressed using a **"matched filter"** operation. The output pulse $u_o(t)$ has the same total energy of the input pulse $s_o(t)$ but is significantly shorter.



Matched filter: Convolution of the received signal $s_0(t)$ with a reference signal $h_0(t)$:

$$u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t-T)\, dT$$

where $\otimes$ stands for convolution

The reference signal $h_o(t)$ (also known as matched filter function) is given by the complex conjugated, time-reverted transmitted chirp $h_0(t) := x_0^*(-t)$ so that $u_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot s_0^*(T-t)\, dT$

Convolution Theorem:

Convolution of $s_0(t)$ **with** $h_0(t)$**:** $u_0(t) = s_0(t) \otimes h_0(t) = \int_{-\infty}^{\infty} s_0(T) \cdot h_0(t - T)\ dT$

FFT convolution is faster than standard (time domain) **convolution**, while producing the same results !

Convolution Theorem:

Time domain		Frequency domain
$u_0(t) = s_0(t) \otimes h_0(t)$	$\leftrightarrow$	$U_0(f) = S_0(f) \cdot H_0(f)$

where: $U_0(f) = \mathrm{FFT}\{u_0(t)\}$ $\quad H_0(f) = \mathrm{FFT}\{h_0(t)\}$ $\quad S_0(f) = \mathrm{FFT}\{s_0(t)\}$

convolution in one domain (time domain) equals multiplication in the other domain (frequency domain).

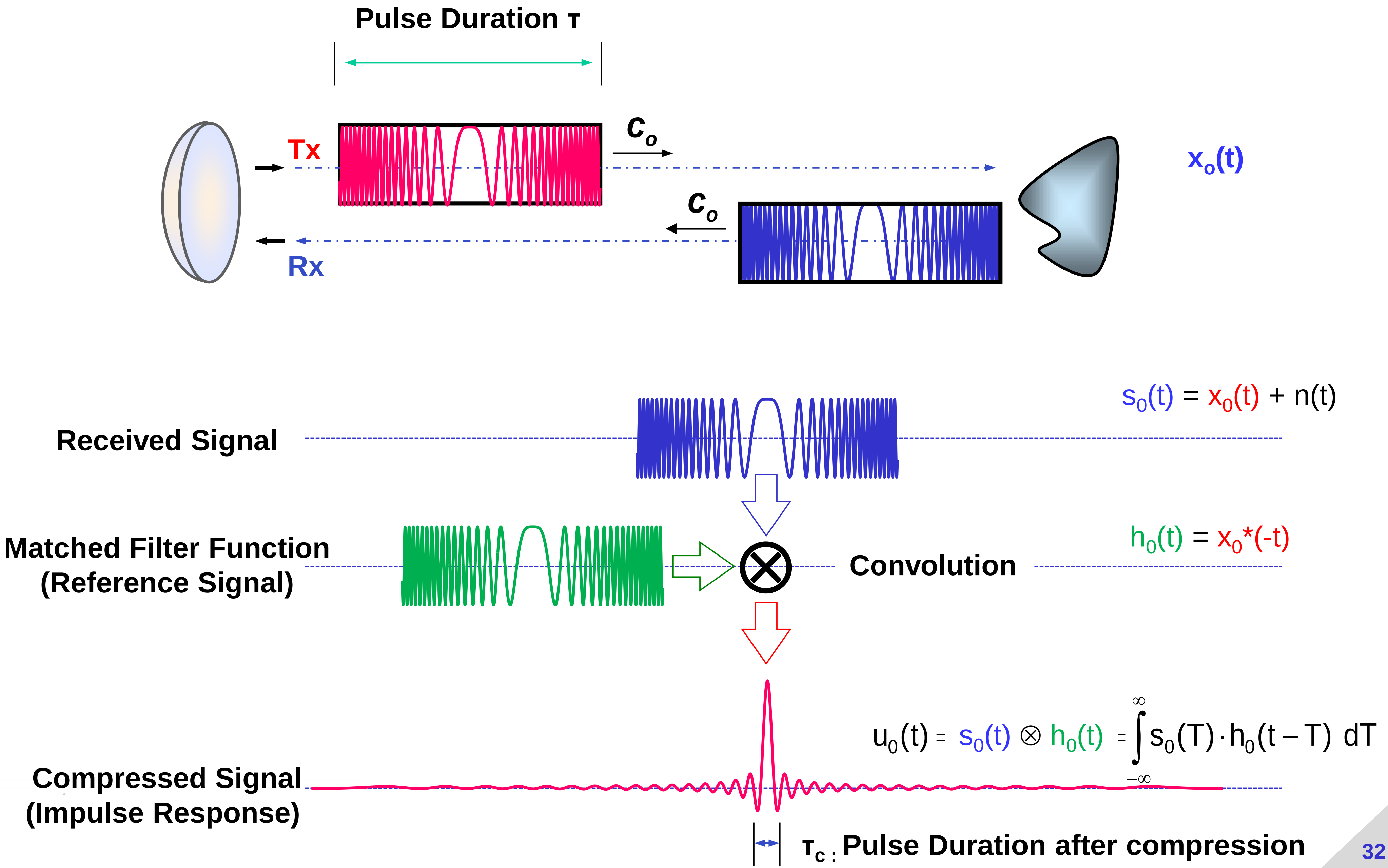
- FFT convolution: N log(N) operations; Time domain convolution: N^2 operations … where N is the filter length.

Convolution of $s_0(t)$ **with** $h_0(t)$**:** $u_0(t) = s_0(t) \otimes h_0(t) = \mathrm{IFFT}\{S_0(f) \cdot H_0(f)\} = \mathrm{IFFT}\{\mathrm{FFT}\{s_0(t)\} \cdot \mathrm{FFT}\{h_0(t)\}\}$

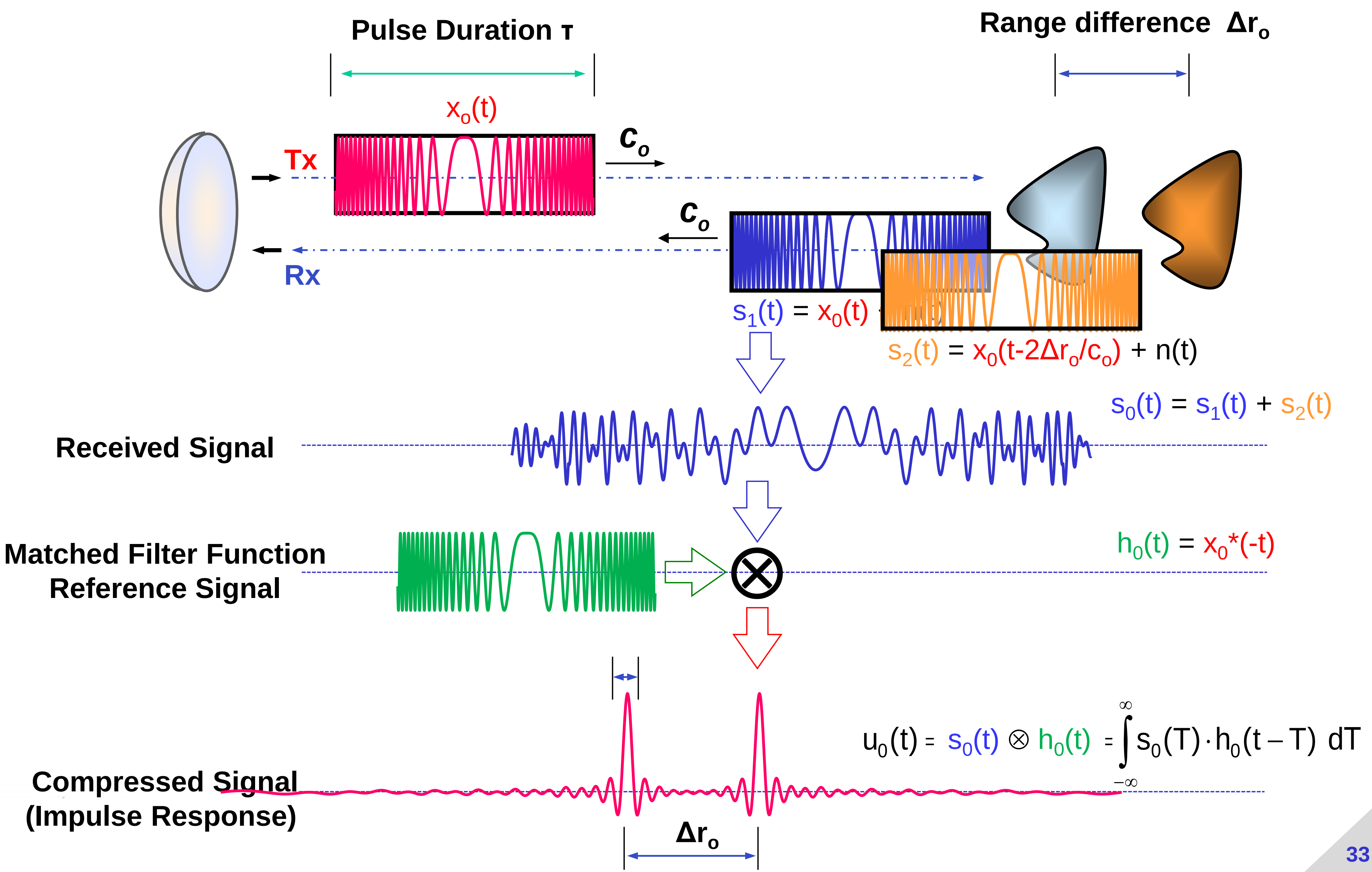
Matched Filter: $h_0(t) = x_0^*(-t)$ $\quad u_0(t) = s_0(t) \otimes x_0^*(-t) = \mathrm{IFFT}\{S_0(f) \cdot X_0^*(f)\} = \mathrm{IFFT}\{\mathrm{FFT}\{s_0(t)\} \cdot \mathrm{FFT}\{x_0(t)\}^*\}$

FFT: Fast Fourier Transform, IFFT: Inverse Fast Fourier Transfrom

Range Resolution



Range Resolution



The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

Part 1: Range Resolution

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Email: kostas.papathanassiou@dlr.de, matteo.pardini@dlr.de, islam.mansour@dlr.de

DLR esa ESAMAAP EEBIOMASS